

Learned static condensation for cut thin-walled TPMS cells with equilibrium correction

[First author]^{a,*}, [Second author]^a

^a[Department, University], [City], [Country]

Abstract

Static condensation of cut thin-walled Schwarz-P-type triply periodic minimal surface cells needs one interior solve for each of their tens of thousands of retained coordinates. We approximate the condensed operator on the complete retained space by neural-initialised condensation with equilibrium correction (NICE): a geometry-conditioned network supplies an initial interior field, and a fixed, geometry-specific multilevel correction reduces its equilibrium residual. As for any admissible extension, the matrix-free condensed stiffness is symmetric, positive semidefinite, reproduces rigid motion and is bounded below by the exact Schur complement; the correction cannot increase its error. Following the interior error through energy participation and the stiffness derivative explains why accurate compliance need not imply accurate local thickness sensitivity. NICE attains a mean directional energy error of 0.074% under consistent tractions on 80 validation geometries not used in training (0.077% on the 60 not used for weight selection; largest geometry mean 0.65%). In two-cell assemblies, its face-load compliance and eight-corner sensitivity errors stay below 0.06% and 0.7% in fourteen configurations of development cells and below 0.28% and 1.5% in eighteen configurations of nine cells outside weight selection, including the five geometries with its largest single-cell errors, whereas the uncorrected network exceeds 3% sensitivity error in four of eleven. In two eight-cell lattices of learned cells, these errors stay within 0.015% and 0.15%. An eight-cell design iteration takes 81–110 s and 8.5–9.2 GiB of GPU memory on one RTX 5090, against 868–1,042 s and 71–86 GiB for the whole-lattice Cholesky solution on 16 CPU threads and 5,864–8,131 s on one. [Placeholder: quantitative

*Corresponding author.

Email address: [corresponding e-mail] ([First author])

outcome of the Section 6.11 design example.]

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1. Introduction

Finite structures built from thin-walled triply periodic minimal surface (TPMS) cells combine spatially varying thickness with supports, external loads and cells cut by the specimen boundary; graded TPMS specimens have been manufactured and mechanically tested Yu et al. (2019). A reusable cell model must respond to the displacement imposed by the surrounding structure. Its returned forces determine the assembled equilibrium and its interior field the local design quantities, so a reduced cell model must approximate both, and its error must be understood in the quantities that analysis and design require.

Static condensation provides such a model without approximation Guyan (1965), Irons (1965): interior equilibrium defines a Schur complement on the retained displacement coordinates, and the equilibrium extension recovers the interior field. For cut thin-walled cells the retained space is large: box-face coefficients carry exterior loads and intercell coupling, and a band of cut-element coefficients carries virtual work on the cut surface, so each cell here retains tens of thousands of coordinates. Forming the Schur complement then needs one interior solve per retained coordinate, and applying it needs an interior factorisation for every geometry. Two approximations reduce this cost. Reducing the retained coordinates, as in port reduction for component-based static condensation Eftang & Patera (2013), Smetana & Patera (2016), changes the displacement patterns that substructures can exchange. Approximating the interior extension on the complete retained space alters only the interior response to each pattern. This is the route of the static-condensation reduced basis element method, with parametric reduced bubble spaces and a posteriori error bounds Huynh et al. (2013), Ballani et al. (2018), and of inexact substructuring, which replaces exact discrete harmonic extensions by approximate ones Dohrmann (2007), Li & Widlund (2007), Klawonn & Rheinbach (2007). We follow the second route with a learned, geometry-conditioned interior extension. For unfitted two-dimensional lattices, BDDC has been combined with reduced-order cell stiffnesses Bonilla Moreno et al.

(2027).

Multilevel iteration supplies the means of correcting such an extension: polynomial smoothing and a Galerkin coarse solve with geometry-fixed coefficients and a prescribed number of steps form a fixed linear map that reduces the interior equilibrium residual at fixed retained displacement Xu (1992). Unlike hybrid neural solvers, which combine learned prediction with relaxation, learned inverse actions or coarse spaces to accelerate one global solve Zhang et al. (2024), Kopaničáková & Karniadakis (2025), Xing et al. (2026), the correction here is a fixed, geometry-specific linear map inside the condensed energy, so that the corrected operator remains symmetric, bounded below by the exact Schur complement and consistent with its complete transpose. A geometry-conditioned network supplies the initial interior field, and the correction completes it for each geometry.

Learned substructures have been developed along a related line. Problem-independent machine learning (PIML), introduced for two-dimensional substructures Huang et al. (2022) and extended to three dimensions Huang et al. (2023), predicts substructure shape functions, imposes rigid-motion constraints and, in its self-consistent form, evaluates the stiffness as $N^T K N$. Its data-free variant trains the shape functions by minimum potential energy without labelled shape functions, extending the displacement of a three-dimensional substructure from 24 corner degrees of freedom Huang et al. (2024). The approach has been extended to Bézier-enriched boundaries Guo et al. (2026a), two-dimensional oversampled bases joined by an overlapping partition of unity Guo et al. (2026b), symmetry-equivariant shape functions Jiang, C. et al. (2026) and isoparametric substructures for complex domains Zhang et al. (2026), and applied to lattice optimisation Xu et al. (2025). Describing each substructure by a few boundary coordinates gives coarse models small enough for very large structures, at the price of a boundary model error independent of the learned interior, which this line of work controls by refining the partition Huang et al. (2024), enriching the interpolation Guo et al. (2026a) or oversampling Guo et al. (2026b). Guo et al. (2026a, Table 1) also retained every boundary node with a learned interior (full node prediction) and set this option aside: for 10^3 -voxel substructures its displacement error, 3.46% against 2.22% with the Bézier boundary, grew, which they attribute to the larger network output, and the denser coupling raised the cost. The present framework retains the complete trace of cut thin-walled cells, 1.7 – 2.6×10^4 coordinates per cell, and restores the accuracy of the learned interior with the equilibrium correction (directional energy error about 0.07%,

assembled compliance errors 0.01–0.06%); its approximation error lies in the interior extension, at the price of tens of thousands of coarse coordinates per cell. Related energy-based models include embedded positive-semidefinite learned elements Parish et al. (2024) and convex neural energy elements, which learn boundary energies from Schur-complement targets and recover linear fields by exact interior lifting Jiang, H. et al. (2026); here one corrected extension defines both the assembled operator and the recovered field.

Three further lines of work bear on this construction. First, component reduced-basis methods build their interior approximation offline from snapshots of a parameter family and certify it a posteriori Huynh et al. (2013); NICE conditions one network on the discrete geometry of each cell, including its cut, and provides no certified bound, although Eq. (10) supplies a computable residual in the norm that governs the stiffness error. Second, learned solver components such as multigrid prolongations Greenfeld et al. (2019), Luz et al. (2020), adaptive FETI-DP coarse spaces Heinlein et al. (2021) and networks trained through the solver Um et al. (2020) accelerate a global solve, and geometry-aware neural operators and graph networks Li et al. (2023a), Li et al. (2023b), Pfaff et al. (2021), multigrid-structured operators He et al. (2024), learned Green's functions Boullé & Townsend (2023) and Ritz-type training E & Yu (2018) learn solution maps; here the learned map is exactly linear in the retained displacement, so that its energy defines an assembled stiffness. Third, in topology optimisation, approximate reanalysis and inexact solves have shown that an accurate objective need not give accurate sensitivities Amir et al. (2009), Amir et al. (2010), Gogu (2015), and coarse multiscale bases with fine-scale correction have been used without scale separation Alexandersen & Lazarov (2015), Efendiev et al. (2013), including numerical shape functions on unfitted coarse elements conditioned against small cuts Chen & Li (2026).

Following the error of this construction to the structural response shows that energy participation can mask the interior error of a weakly participating cell in the compliance while its thickness sensitivity, weighted by the stiffness derivative, remains inaccurate. This is the distinction on which goal-oriented error estimation rests Becker & Rannacher (2001), Oden & Prudhomme (2001), and it has been observed for approximate reanalysis Amir et al. (2009). We quantify it for learned substructures and adopt joint accuracy of compliance and local sensitivity as the design criterion.

Any admissible, rigid-motion-reproducing extension evaluated in the energy form, as in static condensation Guyan (1965), multiscale finite elements

Hou & Wu (1997), component reduced-basis methods Huynh et al. (2013) and learned shape-function substructures when the stiffness is evaluated as $N^T K N$ Huang et al. (2023), yields a symmetric positive semidefinite condensed stiffness with the rigid kernel, bounded below by the Schur complement on the retained trace ($L^T S L$ for a boundary interpolated by L) with a quadratic excess Toselli & Widlund (2005). The correction inherits the energy properties of Chebyshev smoothing and Galerkin coarse projection Golub & Varga (1961), Xu (1992), Xu & Zikatanov (2002). The contributions of this work are threefold. (i) NICE: a geometry-conditioned, matrix-free extension, exactly linear in the retained displacement, on the complete retained space of cut cells in a stabilised cut finite element model, including the cut band, composed with a fixed, geometry-specific multilevel correction inside the energy form together with its complete transpose. (ii) An error analysis for learned substructures that follows the interior error to compliance, through a load-specific participation bound, and to the eight-corner sensitivity, through the stiffness derivative and the amplification of displacement error into energy error in thin walls, and that makes joint accuracy of compliance and local sensitivity a design criterion. (iii) Numerical evidence that NICE meets this criterion on 80 validation geometries, fourteen two-cell configurations and two eight-cell lattices, together with an ablation of the retained representation and a lattice-level cost comparison with the whole-lattice direct solution and conventional exact condensation [Placeholder: and a thickness design, Section 6.11].

Sections 2–5 develop the method and its error relations, Section 6 reports the numerical examples, and Section 7 discusses them and states the limitations.

2. Discrete substructures and static condensation

2.1. Geometry and discrete elastic problem

We consider three-dimensional thin-walled TPMS cells in the reference box $\mathcal{B} = [0, 1]^3$. Their material domain is defined by a Schwarz-P-type trigonometric level-set field, a spatially varying band parameter, and an optional planar cut:

$$\begin{aligned} \phi(x) &= \sum_{a=1}^3 \cos(2\pi x_a), & \tau(x) &= \sum_{c \in \{0,1\}^3} N_c^{Q_1}(x) \tau_c, \\ \Omega(\eta) &= \{x \in \mathcal{B} : |\phi(x)| \leq \tau(x), \mathbf{n} \cdot x \leq b_{\text{cut}}\}. \end{aligned} \tag{1}$$

The eight parameters τ_c , interpolated by the trilinear functions $N_c^{Q_1}$, control the implicit band width and hence the material distribution; they are not pointwise wall thicknesses, and we call them corner thickness parameters and derivatives with respect to them thickness sensitivities. The half-space constraint is omitted for an uncut cell, and η collects geometry, material and discretisation parameters.

The elastic problem uses continuous tensor-product Q_2 displacement functions on a Cartesian background mesh. Integration over the material domain supplies the bulk stiffness, and ghost-penalty stabilisation couples neighbouring active elements to control small-cut effects Burman et al. (2015). The symmetric matrix K defines the stabilised discrete energy $\frac{1}{2}u^T Ku$, and the reference in this work is the equilibrium of this discrete system. Figure 1 shows representative geometries and Table 1 the principal settings; Appendix A gives the integration and assembly details.

2.2. Retained degrees of freedom and equilibrium extension

For each cell, the retained set P contains the active box-face displacement coefficients and all displacement coefficients of active elements carrying a positive-area macro-cut patch. The latter form a cut band: although some of its nodes lie away from the cut plane, their basis functions contribute to displacement and virtual work on that plane. This full retained space, with both intercell and exterior-boundary coordinates, is used throughout prediction and assembly.

Let I denote the remaining internal degrees of freedom, $p = |P|$, $i = |I|$, and $n_a = p + i$. The selectors J_P and J_I extract the two sets from the active displacement vector, and $q = J_P u$ is the retained displacement. Boundary loads act through q ; internal body loads are zero. In the displayed (P, I) ordering, static condensation gives

$$K = \begin{bmatrix} K_{PP} & K_{PI} \\ K_{IP} & A \end{bmatrix}, \quad A = K_{II}, \quad E = \begin{bmatrix} I_p \\ -A^{-1}K_{IP} \end{bmatrix}, \quad (2)$$

$$S = K_{PP} - K_{PI}A^{-1}K_{IP} = E^T K E.$$

We assume $K \succeq 0$ and $A \succ 0$, so prescribing q removes every internal zero-energy motion. The equilibrium displacement extension $u = Eq$ uniquely minimises the discrete energy over $J_P u = q$, and Sq is its work-conjugate retained force. For a connected free substructure, we additionally assume that K has exactly six rigid-body modes whose restriction to P has rank

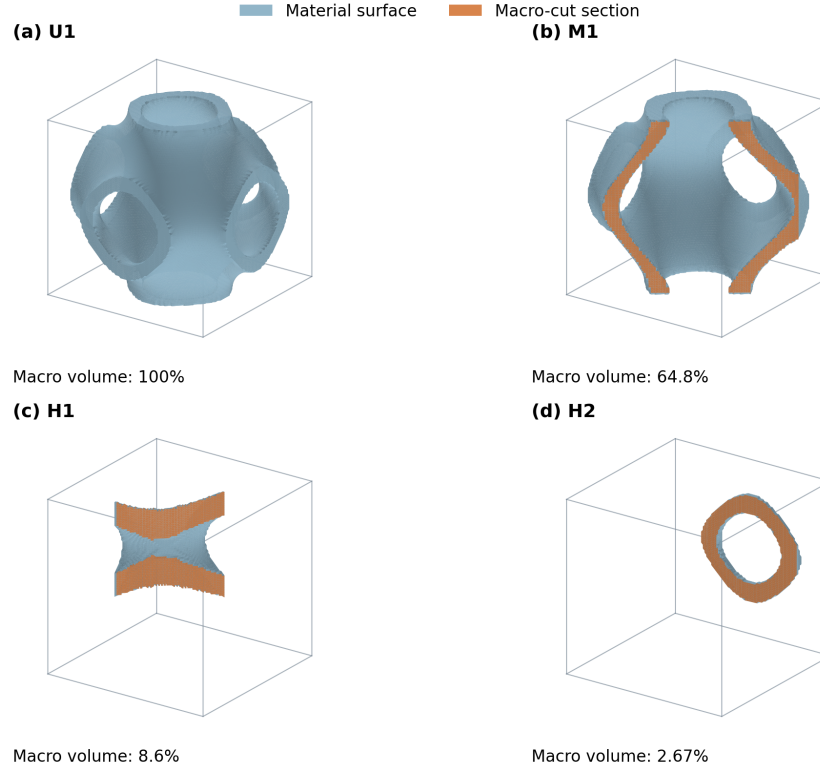


Figure 1: **Representative validation geometries.** The same unit-box scale and viewing direction are used for (a) uncut U1, (b) moderately cut M1, and (c,d) heavily cut H1 and H2. Blue denotes the material surface and orange the macro-cut section. Percentages indicate the retained macro-domain volume relative to the unit box, before intersection with the thin-wall material. Surfaces are reconstructed from Eq. (1) (Supplementary Note S1).

six. Geometry generation enforces this: a cell is accepted only if its active background elements form a single face-connected set carrying retained box-face coordinates (10 of the 1,109 cells generated for the training expansion were rejected), and every training, validation, neighbour and lattice cell used here passes the check. The check establishes connectivity of the stabilised discrete model, in which material connected only through partially filled elements is coupled through shared basis functions and the ghost penalty; the deployed operator reproduces the six rigid-body modes to a relative energy of 2×10^{-11} (Section 6.2).

2.3. Assembly and response measures

For substructure m , the Boolean assembly map B_m extracts $q_m = B_m U$ from the free global retained displacement U . Coincident box-face coordinates, identified by their background-grid position, are shared; retained cut-band coordinates outside the box faces remain local to their substructure. Because the thickness field on a shared face depends only on its four corner values and ϕ is periodic, the material patches of two neighbours coincide on every face not intersected by a cut; where a cut removes material from one side only, the unmatched face coordinates remain coordinates of the cell that carries them. Fixed homogeneous supports are incorporated into B_m . The reference and approximate assembled systems, which use the same maps B_m , are

$$\mathbb{K} = \sum_m B_m^T S_m B_m, \quad \hat{\mathbb{K}} = \sum_m B_m^T \hat{S}_m B_m, \quad \mathbb{K}U = f_g, \quad \hat{\mathbb{K}}\hat{U} = f_g. \quad (3)$$

Here $\hat{S}_m$ is the approximate condensed stiffness of Section 3. Local condensation commutes with assembly because the internal sets are disjoint, and the supported reference matrix $\mathbb{K}$ is assumed positive definite.

Under a prescribed force f_g , compliance is $C = f_g^T U$, with $\hat{C} = f_g^T \hat{U}$ and relative error $e_C = |\hat{C}/C - 1|$. Local operator accuracy is measured by the relative directional energy excess of Section 4.1. Thickness sensitivities use the eight-component vector of Section 4.3, with relative Euclidean error $e_s = \|\tilde{\mathbf{s}} - \mathbf{s}\|_2 / \|\mathbf{s}\|_2$. All relative measures use a nonzero reference denominator.

Table 1: Discretisation and internal-correction settings.

Quantity	Setting
Reference geometry	Unit box; Schwarz-P-type implicit band with trilinear corner parameters and an optional planar cut
Displacement approximation	Continuous tensor-product Q_2 solid elements on a Cartesian background mesh
Background resolution, validation geometries	$n = 32$ elements per axis; 65 Q_2 node positions per axis
Isotropic material, validation geometries	Normalised Young's modulus $E_Y = 1$, Poisson's ratio $\nu = 0.3$
Ghost-penalty coefficient, validation geometries	$\gamma = 10^{-4}$
Retained space	Active box-face coefficients and all coefficients of active elements carrying a positive-area macro-cut patch
Standard volume integration	4^3 initial subcells; one local refinement of partial subcells; clipped Kuhn tetrahedra with quadrature-rule parameter 4
Smoothing interval	$[b/30, b]$, with $b = 1.05\hat{\lambda}_{\max}$ estimated by 40 power iterations
Principal interior coarse space	Trilinear vector functions on a 17^3 -vertex grid, restricted to the internal degrees of freedom
Thickness-difference step	$h_c = 10^{-5}\tau_c$, with fixed active coordinates and ghost contribution
Correction of NICE	8 Chebyshev steps, $Q_1(17)$ Galerkin solve, 8 Chebyshev steps; same sequence in training and evaluation

Quantity	Setting
Arithmetic	Network, rigid reconstruction and retained entries in single precision (true fp32 convolutions, no TF32); stiffness actions of the condensed product and energies in double precision; correction (smoothing and coarse solve) in double precision in the accuracy studies of Sections 6.2–6.9 and in single precision in the timed route of Table 5, with the corrected field returned to single precision (Appendix F.3)

The mesh, material and stabilisation entries apply to all 80 validation geometries; lengths are relative to the unit box and the modulus is normalised. Coarse dimensions and smoothing counts of other correction sequences are reported with their comparisons.

3. Geometry-conditioned neural displacement extension

Figure 2 shows how the learned extension and the internal equilibrium correction define both the condensed action and the displacement recovered after assembly.

3.1. Geometry-conditioned multilevel displacement extension

The learning target is the internal equilibrium map $q \mapsto E_I q = -A^{-1}K_{IP}q$, whose approximation provides both a displacement field and, through its energy, the condensed operator on the complete retained space. The architecture must adapt to the material and support of each cell and respect superposition at fixed geometry, which leads to a nonlinear geometry branch coupled to a linear, multilevel displacement branch (Figure 3).

Rigid motion is separated before learning. Let $R \in \mathbb{R}^{n_a \times 6}$ contain three translations and three rotations about a common centre, and let $R_P = J_P R$. The coefficient extractor $C_R = (R_P^T R_P)^{-1} R_P^T$ and projector $\Pi_P = I_p - R_P C_R$ decompose the input into rigid coefficients $C_R q$ and retained deformation $q_d = \Pi_P q$. The network extends only q_d , so rigid reconstruction is independent of training accuracy.

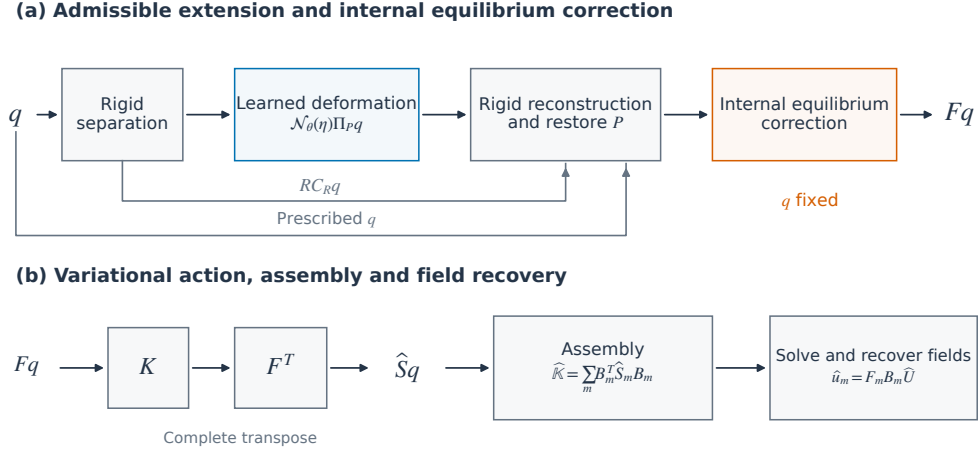


Figure 2: **Learned displacement extension, equilibrium correction and variational assembly.** (a) Rigid motion is separated from the retained displacement before the deformation is extended by the network. The rigid field is reconstructed and the prescribed retained values are restored; correction then reduces internal imbalance at fixed retained displacement. (b) Applying the local stiffness and the complete extension transpose gives the work-conjugate retained force. The assembled solution supplies the inputs for local field recovery. The neural extension is detailed in Figure 3; Section 5 defines the internal correction.

In the geometry branch, element moments resolve the material distribution within each background cell, and node features identify retained and cut-band membership, weak support, local stiffness magnitude and position. Encoders form element and node embeddings, which exchange information in two rounds; coefficient heads then assign weights to prescribed element and ghost-face incidences, to transfers between latent grids and to coarse-grid convolutions. These coefficients are fixed for all displacement directions on the same geometry.

The displacement branch lifts the three components of q_d to 32 channels on retained nodes, with the internal features set to zero, defining $X^0(q)$. Local interactions propagate these values through the element and ghost-face neighbourhoods: each gathers the features on a 27-slot stencil into four weighted heads, mixes the channels within each head and scatters the result back to the incident nodes. With internal and retained masks M_I, M_P acting on nodal rows, an interaction updates the latent field by

$$X^{\ell+1} = M_I \left[X^\ell + \sum_h \mathcal{S}_{\ell h}(\eta) (\mathcal{G}_{\ell h}(\eta) X^\ell W_{\ell h}) \right] + M_P X^0(q). \quad (4)$$

Here $\mathcal{G}_{\ell h}$ and $\mathcal{S}_{\ell h}$ are geometry-weighted gather and scatter maps, $W_{\ell h}$ mixes channels, and the final term restores the retained features. A U-shaped latent hierarchy through grids with 65, 33, 17 and 9 positions per axis supplies longer-range communication, with residual convolutions on each coarser level and additive skips on the upward pass. Four local interaction pairs precede the hierarchy, four follow it, and four further pairs act on stencils containing weakly supported nodes; a linear 32-to-3 map reconstructs nodal displacement. Appendix G gives the complete order of operations and coefficient definitions.

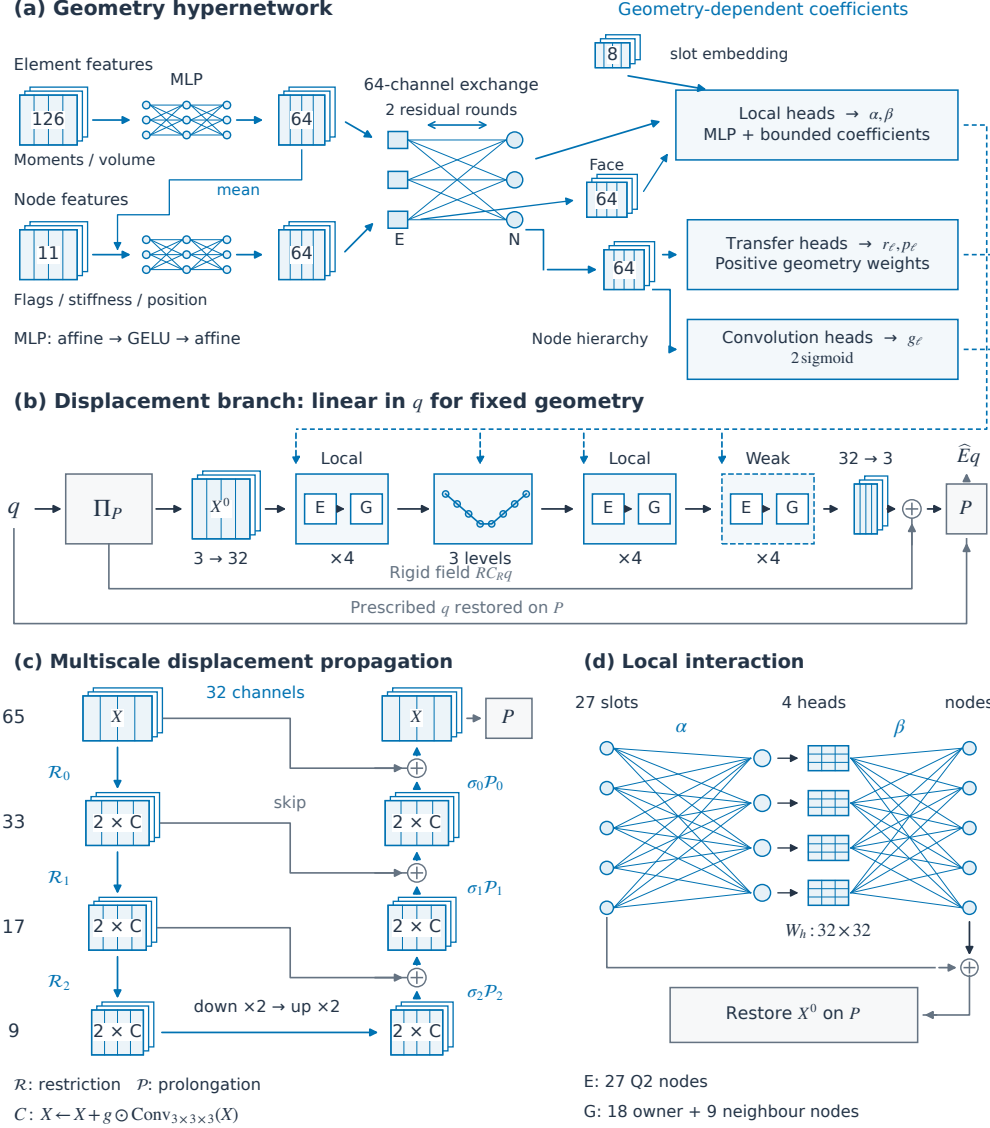
Every displacement operation is linear, and the nonlinear encoders act only on geometry, so the raw map $\mathcal{N}_\theta(\eta)$ obeys superposition at fixed η . Adding back the rigid field and restoring the original retained values gives

$$\hat{E}q = J_P^T q + J_I^T J_I [RC_R q + \mathcal{N}_\theta(\eta) \Pi_P q]. \quad (5)$$

Consequently, $J_P \hat{E} = I_p$ and $\hat{E} R_P = R$: admissibility and rigid reproduction hold by construction.

3.2. Variational condensed stiffness

The learned field becomes a mechanical operator through its energy with the original stiffness. Let $\mathcal{W}$ be a fixed linear internal correction that pre-



Training: update θ through both branches.

Inference: fix θ ; cache geometry coefficients.

Figure 3: **Geometry-conditioned neural displacement architecture.** (a) Element and node encoders produce 64-channel geometry embeddings, followed by two rounds of residual exchange; coefficient heads condition the local interactions, grid transfers and convolutions. (b) The displacement branch propagates the nonrigid retained input through four local interaction pairs, the multilevel block, four further local pairs and four pairs on weakly supported stencils; linear maps connect the three displacement components to 32 latent channels, and deterministic bypasses reconstruct rigid motion and restore the retained values. (c) Restriction and prolongation connect grids with 65, 33, 17 and 9 background positions per axis; each coarse level has two residual convolutions on each pass. (d) A local interaction uses geometry-weighted gathering, four channel-mixing heads and scattering, followed by residual addition and retained-value restoration. E and G denote element and ghost-face interactions. Blue dashed arrows carry geometry-dependent coefficients; solid arrows carry features or displacement states. For fixed geometry, the complete displacement path is linear.

serves retained values, with the identity corresponding to no correction. For each geometry, the correction schedule and its coefficients are fixed independently of the displacement query. The final extension and its condensed stiffness are

$$F = \mathcal{W}\hat{E}, \quad \hat{S} = F^T K F, \quad \hat{\mathcal{U}}(q) = \frac{1}{2} q^T \hat{S} q, \quad \hat{f}_P = \hat{S} q. \quad (6)$$

For a virtual retained displacement δq , the corresponding full virtual field is $F\delta q$, and $\delta\hat{\mathcal{U}} = (F\delta q)^T K F q = \delta q^T F^T K F q$. Applying the condensed operator therefore requires the transpose of the complete extension after the stiffness action, including the rigid reconstruction, retained-value restoration and any internal corrections. With $F_I = J_I F$, its block form is

$$\hat{S} q = (K F q)_P + F_I^T (K F q)_I. \quad (7)$$

The second term transfers the work of the internal residual to the retained coordinates. It vanishes for an equilibrated field; otherwise it is needed for the returned force to be the derivative of the stated energy. Symmetry and positive semidefiniteness follow from $K = K^T \succeq 0$, without requiring symmetry of the learned gather, scatter or grid-transfer maps. Under the rigid-kernel assumption of Section 2.2 and $F R_P = R$, the only null modes of $\hat{S}$ are the retained rigid-body modes (Appendix B); Appendix E derives the complete transpose, and Appendix J.2 shows why it makes the operator error quadratic. After an assembled solve, $F_m B_m \hat{\mathcal{U}}$ supplies the local reconstructed displacement for response and sensitivity evaluation.

3.3. Training directions and objective

Training probes the extension through retained displacement directions: prescribed polynomial and multiscale displacements supply broad spatial content, and responses to equilibrated nodal loads, consistent tractions, spring supports and neighbouring cells supply mechanically generated directions. Rigid components are removed and each direction is normalised to $q_j^T S q_j = 1$ with the reference solution. For a batch of B directions, the objective is

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{j=1}^B \log(q_j^T \hat{S} q_j) + w_s \frac{1}{|\mathcal{J}_s|} \sum_{j \in \mathcal{J}_s} \frac{\|\tilde{\mathbf{s}}_j - \mathbf{s}_j\|_2^2}{\|\mathbf{s}_j\|_2^2}. \quad (8)$$

The energy term is the mean logarithm of the predicted-to-reference energy ratio. For an admissible extension, the variational identity of Section 4 makes its minimum correspond to the equilibrium field on each sampled direction, a Ritz principle. The sensitivity term compares the complete eight-corner field-based sensitivity vectors on the set $\mathcal{J}_s$ of directions with reference sensitivity labels; the reported configurations use $w_s = 1$, and the effect of this term was not isolated. Gradients with respect to θ pass through the reconstructed field to the geometry encoders, coefficient heads and linear displacement maps. Directions with large energy ratios, found by a block search on the rigid complement, are added during training, and geometry augmentation uses the 48 cube symmetries (Appendix G). For NICE the objective is evaluated on the corrected extension (Section 5.4); Table 2 identifies the predictors and Table ST01 records the training and weight-selection settings.

4. Extension error and mechanical response

With the retained coordinates fixed, the local approximation lies in the interior field. We relate its departure from equilibrium to the condensed stiffness and examine how assembly and thickness differentiation weight it. Appendix J.1 collects the assumptions. Several relations are classical and are restated in the present notation: the energy minimality of the discrete harmonic extension and the Ritz identity (9) Fraeijs de Veubeke (1965), Toselli & Widlund (2005); the residual form (10) and the residual-corrected functional (18) Becker & Rannacher (2001); the energy orthogonality behind Eq. (11); the self-adjoint compliance sensitivity Haftka & Gürdal (1992), Bendsøe & Sigmund (2003); and, in Section 5, Chebyshev semi-iteration Golub & Varga (1961) and the two-grid error operator Hackbusch (1985), Trottenberg et al. (2001), Xu & Zikatanov (2002), Falgout et al. (2005). Derived here for learned substructures are the participation bounds of Eq. (12) and Appendix J.3, the sensitivity bound of Appendix J.4, the order of the complete surrogate derivative in Appendix J.5 and the identity $\varepsilon = \delta^2 \kappa$ used in Section 6.4.

4.1. Variational energy error

For the specified symmetric stiffness $K \succeq 0$, with $A = K_{II} \succ 0$ and zero internal body loads, Eq minimises the energy at prescribed q . Any admissible linear extension F , with $J_P F = J_P E = I_p$, differs from it only internally.

Writing $H = J_I(F - E)$, internal stationarity $J_I K E = 0$ eliminates the cross terms in $F^T K F$ and gives the discrete Ritz identity

$$\boxed{\hat{S} - S = H^T A H \succeq 0.} \quad (9)$$

The learned extension therefore adds stiffness in proportion to the energy of its internal error; the absence of a first-order term follows from equilibrium of the reference field. For $u = Eq$, $\hat{u} = Fq$ and $d_I = Hq$, the internal residual $r_I = (K\hat{u})_I$ satisfies $r_I = Ad_I$. The relative directional energy excess $\varepsilon(q) = q^T(\hat{S} - S)q/(q^T S q)$ can consequently be written as

$$\varepsilon(q) = \frac{d_I^T A d_I}{q^T S q} = \frac{r_I^T A^{-1} r_I}{q^T S q}. \quad (10)$$

The squared A^{-1} -norm of the internal residual, available without the reference extension, is thus the stiffness-error measure; Appendix B gives the all-direction norm, which a finite direction bank samples.

4.2. Assembled compliance and energy participation

Consider the exact equilibria of the two supported systems in Eq. (3), with the same local matrices, assembly maps and homogeneous supports, and the same nonzero retained load. Writing $u_m = E_m B_m U$, $\hat{u}_m = F_m B_m \hat{U}$ and $a(v, v) = \sum_m v_m^T K_m v_m$, global equilibrium and local energy orthogonality give (Appendix C)

$$\boxed{C - \hat{C} = a(\hat{u} - u, \hat{u} - u) = \|\hat{U} - U\|_{\mathbb{K}}^2 + \sum_m \|H_m B_m \hat{U}\|_{A_m}^2.} \quad (11)$$

Compliance underestimation is therefore the total reconstructed error energy, with orthogonal contributions from the changed retained solution and from the internal departure from equilibrium at that solution.

The influence of a local approximation depends on how much energy that substructure carries under the applied load. At the exact assembled traces $q_m = B_m U$, define $w_m = q_m^T S_m q_m / C$ and $\beta = \sum_m w_m \varepsilon_m(q_m)$, with contributions of rigid zero-energy responses defined as zero. The energy fractions w_m sum to one, and

$$0 \leq \frac{C - \hat{C}}{C} \leq \frac{\beta}{1 + \beta} \leq \beta. \quad (12)$$

Small participation can thus attenuate a substructure's effect on compliance even when its local field remains inaccurate (Appendix J.3).

4.3. Field-based sensitivity and the complete design derivative

On a differentiable design interval with fixed active and retained coordinates, assembly maps, homogeneous supports and a design-independent load, the exact compliance sensitivity is $s_c = -u^T K_{,c} u$, where $K_{,c} = \partial K / \partial \tau_c$; contributions are summed over affected substructures. Internal stationarity gives $S_{,c} = E^T K_{,c} E$, eliminating the design derivative of the exact extension from the force-controlled compliance derivative Giles & Pierce (2000), Haftka & Gürdal (1992).

The field-based sensitivity estimate $\tilde{s}_c = -\hat{u}^T K_{,c} \hat{u}$ uses the same stiffness derivative with the reconstructed field. At the same retained displacement,

$$\tilde{s}_c - s_c = -2d^T K_{,c} u - d^T K_{,c} d, \quad u = Eq, \quad d = Fq - Eq. \quad (13)$$

Internal equilibrium sets $(Ku)_I = 0$ but generally leaves $(K_{,c}u)_I \neq 0$, so the linear cross term survives, and decreasing energy error need not decrease sensitivity error monotonically. For corner thickening, nonnegative trilinear shape functions generate nested material domains, and with fixed basis, material and ghost contribution the exact integral gives $K_{,c} \succeq 0$ (Eq. J.6); a numerical derivative inherits this only while its quadrature preserves the nesting of the material domains, and the discrete moments used here violate it in a few elements of the cut cells (Appendix J.5, Supplementary Table ST15). Even then the cross term can have either sign (Appendix J.4).

The derivative of the surrogate compliance also contains the design dependence of the extension. For a parameter affecting one substructure, differentiation at fixed retained coordinates gives

$$\hat{C}_{,c} = -\hat{u}^T K_{,c} \hat{u} - 2\hat{q}^T F_{,c}^T K \hat{u} = \tilde{s}_c - 2(F_{I,c} \hat{q})^T r_I, \quad \hat{u} = F\hat{q}. \quad (14)$$

Here $\hat{q}$ is the surrogate's assembled retained solution and $r_I = (KF\hat{q})_I$; affected-substructure contributions are summed. The second term couples the extension's design dependence to its internal imbalance and vanishes for an equilibrated extension. All sensitivities reported in this paper are the field-based estimates $\tilde{s}_c$, the first term: they estimate the exact sensitivity and are not the derivative of the surrogate compliance. Energy accuracy alone does not control the complete derivative (Appendix J.5, Section 7.3); Appendix H specifies the numerical stiffness derivatives.

These derivatives hold on intervals where the active elements, ghost faces, retained coordinates, the network's binary node features and the coarse-factor shift are fixed. Across such a switch the discrete reference compliance

itself can jump, a non-smoothness shared by exact condensation on the same background mesh; because $0 \leq C - \hat{C} \leq \beta C$ at every design (Eq. 12), any jump that the surrogate adds to that of the reference is bounded by its own error level.

5. Internal equilibrium correction

Equations (10) and (11) suggest reducing the internal imbalance of the learned field at fixed retained motion; polynomial relaxation and coarse energy minimisation address different components of it.

5.1. Error spectrum and polynomial relaxation

Set $D = \text{diag}(A)$ and order the generalised eigenvectors by increasing eigenvalue, $Av_j = \lambda_j Dv_j$, with $v_j^T Dv_k = \delta_{jk}$. For $c_j = v_j^T Dd_I$,

$$d_I^T Ad_I = \sum_j \lambda_j c_j^2, \quad \chi_\ell(q) = \frac{\sum_{j=1}^\ell \lambda_j c_j^2}{d_I^T Ad_I}. \quad (15)$$

The fraction χ_ℓ locates the error energy in the first ℓ Jacobi-scaled internal modes (the exact-field fraction uses $u_I^T Au_I$).

The learned field is an initial approximation to $Au_I = -K_{IP}q$, improved by Jacobi-preconditioned Chebyshev semi-iteration with q fixed; a prescribed iteration count and geometry-dependent coefficients keep the corrected extension linear. Polynomial relaxation supplies a parallelisable component of multigrid methods Adams et al. (2003), the combination of a trial extension with smoothing is the principle of smoothed-aggregation prolongation Vaněk et al. (1996), and pairing relaxation with a learned initial field exploits their complementary error-reduction properties Zhang et al. (2024).

A degree- k error polynomial maps d_I to $P_k d_I$, $P_k = p_k(D^{-1}A)$, multiplying each coefficient in Eq. (15) by $p_k(\lambda_j)$. The Chebyshev polynomial targets $[a, b]$, $0 < a < b$, and is nonexpansive in the A -energy norm if the actual positive spectrum of $D^{-1/2}AD^{-1/2}$ lies in $(0, b]$. Modes below a can decay slowly, which motivates a complementary coarse update. The correction sequences use $a = b/30$, with b from a power estimate increased by 5%; Section 6.2 reports its verification, and Appendix D gives the polynomial, recurrence, interval estimator and contraction bound.

5.2. Interior coarse correction

Let $V \in \mathbb{R}^{i \times n_c}$ have full column rank, $A_c = V^T A V$, and $b_r = V^T r_I$. Minimising the energy over $u_I + \text{range } V$ gives

$$u_I^c = u_I - V A_c^{-1} b_r, \quad d_I^c = (I_i - V A_c^{-1} V^T A) d_I, \quad \|d_I\|_A^2 - \|d_I^c\|_A^2 = b_r^T A_c^{-1} b_r. \quad (16)$$

The Galerkin solve enforces equilibrium against the coarse variations, and the last equality measures the energy removed by that projection Xu (1992). The principal basis consists of trilinear vector functions on a 17^3 -vertex grid restricted to internal coordinates (Appendix F); all coarse spaces preserve the full retained space.

Placing the coarse update between two k -step smoothing stages combines these corrections. With $C_V = I_i - V A_c^{-1} V^T A$ and $H_0 = J_I(\hat{E} - E)$, the resulting error is

$$H_{\text{tg}} = P_k C_V P_k H_0, \quad \hat{S}_{\text{tg}} - S = H_{\text{tg}}^T A H_{\text{tg}}. \quad (17)$$

For the exact Galerkin update and $\|P_k\|_A \leq 1$, $S \preceq \hat{S}_{\text{tg}} \preceq \hat{S}_0$: the correction cannot increase the error in the A -energy norm, and by Eq. (11) it moves exact assembled compliance toward the reference, although it does not order the sensitivity error in Eq. (13) (Appendix J.8; Appendix J.9 gives a matrix example in which the energy error falls while the sensitivity error rises). This ordering is guaranteed, whereas the size of the reduction is not. For the spectra of the detailed cells the polynomial estimate of Appendix D guarantees little more than non-expansion and uses no approximation property of the coarse space, so the reductions reported in Section 6.5 are empirical; repeating the cycle cannot increase the energy error, whereas changing the smoothing degree changes the polynomial and is not guaranteed to be monotone.

5.3. Energy-consistent operator application

Geometry-dependent coefficients, maps, smoothing interval and coarse factorisation are prepared once per geometry and reused across retained inputs.

Algorithm 1. Corrected condensed stiffness action.

1. Extend q with $\hat{E}$ and apply the prescribed pre-smoothing, coarse and post-smoothing sequence at fixed retained values to obtain $u = Fq$.

2. Form $y = Ku$ and return $F^T y$: transpose the complete correction in reverse order, followed by the transpose learned extension.
3. Assemble these actions through Eq. (3); after the global solve, recover $F_m B_m \hat{U}$ for field and sensitivity evaluation.

Each forward cycle uses $2k$ smoothing steps and one coarse solve, with the corresponding reverse operations in the transpose (Appendices E–F).

At the assembled level, the energy relation also separates approximation error from incomplete equilibrium. For an approximate solution $\bar{U}$, define the recomputed residual $\rho = f_g - \hat{\mathbb{K}}\bar{U}$ using the stated variational operator and the recovered fields $\bar{u}_m = F_m B_m \bar{U}$. Then

$$C - f_g^T \bar{U} = a(\bar{u} - u, \bar{u} - u) + \bar{U}^T \rho. \quad (18)$$

The signed residual work $\bar{U}^T \rho$ determines how an incomplete solution affects the compliance comparison. Appendix J.6 gives the residual-corrected functional and the additional consistency term required when the numerical action differs from the energy operator.

5.4. Training on the deployed operator

NICE is trained on the operator that is deployed: Eq. (8) is evaluated on the corrected extension $F = \mathcal{W}\hat{E}$, with the 8 / $Q_1(17)$ / 8 sequence of Section 5.2. Because $\mathcal{W}$ is linear and fixed for a given geometry, the gradient passes through the smoothing recurrence and the coarse solve to the network parameters, as in solver-in-the-loop training Um et al. (2020). The smoothing interval and coarse factorisation depend only on K and are recomputed per geometry, so no trainable parameters are added.

6. Numerical examples

The examples follow the error of the learned extension from single cells, through its correction, to assembled compliance and local thickness sensitivity, and close with an ablation of the retained representation and a cost comparison. All errors are measured against equilibrium of the discrete problem of Section 2.

6.1. Geometries, predictors and loading conditions

The 80 validation geometries comprise 20 uncut, 20 lightly, 20 moderately and 20 heavily cut cells, with uniform, affine or mixed trilinear thickness fields and corner parameters from 0.1762 to 0.6983 (Table ST02). Each carries at most one planar cut with normal $(\cos \vartheta, \sin \vartheta, 0)$, $0 < \vartheta < \pi/4$; heavy cuts retain less than one third of the macro-box volume, moderate cuts between one and two thirds, and light cuts more than two thirds. U, L, M and H identify the uncut, lightly, moderately and heavily cut cells used for detailed comparisons (Supplementary R1).

Five predictors are compared (Table 2). The base network was trained without correction. Three networks continue it for 15,000 updates: with the complete correction of Section 5.2 inside the training loop (NICE, Section 5.4), with eight smoothing steps and no coarse solve in every training step (Smoothing-trained), or without correction (Uncorrected). NICE-post is the base network with NICE's correction $\mathcal{W}$ applied at deployment, without further training. Two predictors from earlier comparisons appear in the supplement only (Supplementary R1).

Table 2: Predictors compared in the numerical examples.

Predictor	Network weights	Training pool (geometries)	Correction in training	Correction at evaluation
NICE	Base network continued for 15,000 updates	591	8 / $Q_1(17)$ / 8	8 / $Q_1(17)$ / 8
NICE-post	Base network	305	None	8 / $Q_1(17)$ / 8
Smoothing-trained	Base network continued for 15,000 updates	591	8 smoothing steps	8 smoothing steps
Uncorrected	Base network continued for 15,000 updates	591	None	None

Predictor	Network weights	Training pool (geometries)	Correction in training	Correction at evaluation
Base network	40,000 updates, selected at 30,000	305	None	None

All predictors share the architecture of Section 3 (603,409 trainable parameters and 55 fixed coefficient bounds; Table ST20). The continuations draw from a pool of 591 geometries, 304 of the base network's 305 and 287 produced later (one near-mechanism cell was removed); with one pool replacement every 100 updates, each visits about 150 distinct geometries, in the same order for all three. Measured to the last update on one NVIDIA GeForce RTX 5090, the base network's training took 4.6 h and NICE's continuation 3.5 h (5.0 h and 4.8 h including all selection evaluations; NICE's run was interrupted at update 5,600 and resumed from update 4,000, and both segments are logged; peak device memory 29.6 GiB). Generating the direction banks and exact sensitivities for the 691 training and validation geometries took about 42 GPU-hours and occupies 1.05 TB, so NICE's lineage, data included, cost about 60 GPU-hours offline (Table ST01).

Twenty of the 80 validation geometries (6 uncut, 14 cut) entered weight selection: each continuation was scored at 7,500 and 15,000 updates and the final weights were retained in every recorded case, the base network's weights were selected among four checkpoints, and NICE was designated the principal predictor after all predictors had been compared on the 80 geometries and the pairs. Statistics are therefore also given for the 60 geometries outside selection. All cells of the assembly examples belong to the 20 selection geometries and served repeatedly as development cases; nine further cells outside weight selection are assembled in Section 6.6 as an independent check. Population statistics average the directional energy excess within each geometry and loading class and weight geometries equally (Appendix A.2), so the population maximum is the largest geometry mean.

The assembly examples join a learned target to an exact, uncut neighbouring cell whose thickness is continuous across the interface (Figure 4); where the target's cut plane meets the shared face, the pair is a diagnostic assembly rather than a physically cut specimen. Each configuration has six face loads, the three Cartesian traction directions applied separately to the

target and neighbour faces, and compliance and each cell's eight-parameter sensitivity vector are compared with their exact counterparts, with the nodal load fixed at its base-design value. We use 3% on compliance and on each cell's sensitivity vector as a common visual reference, taking maxima over the six loads and, for sensitivity, both cells; the threshold is not derived from an optimiser tolerance. The directions of the sensitivity vectors are reproduced more closely than their magnitudes: in the fourteen development configurations, NICE's eight-corner vectors have cosines of at least 0.999998 with the exact ones (Supplementary Note S6.3). Three tractions on a target cut face are considered separately.

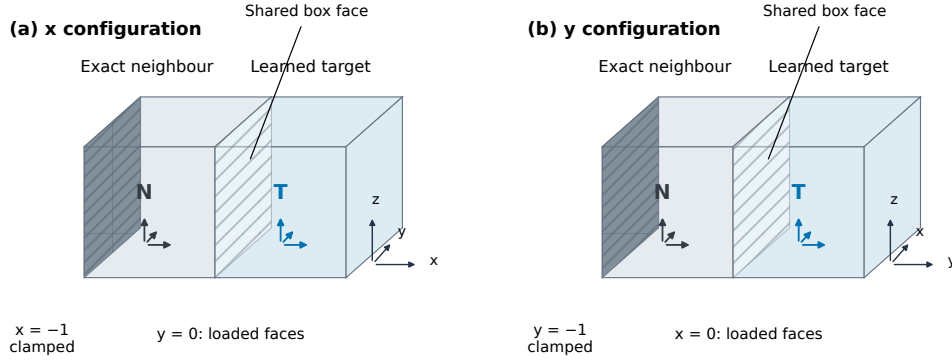


Figure 4: **Supports and loading of the two-cell examples.** (a) Configuration x: the neighbour is translated by $(-1, 0, 0)$, the face $x = -1$ is clamped, and face tractions act at $y = 0$. (b) Configuration y: the translation is $(0, -1, 0)$, the face $y = -1$ is clamped, and tractions act at $x = 0$. Each target (T) and neighbour (N) face carries separate x-, y- and z-directed consistent-traction loads. Coincident box-node coordinates are shared across the interface; non-box cut-band coordinates remain local. Cut targets also receive three macro-cut tractions, analysed separately.

6.2. Verification of the discrete reference and of the corrected operator

To verify the discrete reference, single cells clamped on one box face and loaded by a unit consistent traction on another were solved at $n = 24$ to 48 (Figure S01, Supplementary Note S2), and H1, H2 and two further validation cells up to $n = 64$. For M1, M2 and U1, the compliance at $n = 32$ differs from the finest level by at most 0.057% and the eight-corner

sensitivities by at most 0.11%. The heavily cut H1, clamped on $x = 0$ and loaded on $y = 0$ because its retained part carries no material on its z -faces, differs from $n = 48$ by up to 0.99% in compliance, for the load normal to the loaded face, and by 0.97% in sensitivity. Refinement to $n = 64$ (Supplementary Table ST14) does not converge monotonically for H1: its compliance at $n = 48$ and 56 exceeds the $n = 64$ value by up to 1.1% and 1.4%, while $n = 32$ happens to lie within 0.07% of it. The heavily cut H2 converges monotonically but has not converged at $n = 64$: at $n = 32$ its compliance deviates by up to 0.93% and its sensitivity vector by up to 3.6%, and at $n = 48$ still by 0.27% and 1.2%. For the geometry with NICE's largest single-cell error and for the thinnest-walled validation cell, $n = 32$ differs from $n = 64$ by at most 0.14% and 0.28% in compliance and 0.23% and 0.44% in sensitivity. The discretisation error of the $n = 32$ references is therefore 0.1 to 0.3% in compliance for most cells, but of order one percent in compliance and several percent in sensitivity for heavily cut ones. The comparisons therefore measure the approximation of the discrete operator, not of the continuum. This discrete model also defines the design problem, so a coarser reference would change the model rather than the cost of reaching it; the surrogate's energy error lies well below the reference's discretisation error in heavily cut cells, whereas its largest assembled sensitivity errors are comparable to the reference's sensitivity change under refinement. Varying the penalty coefficient between 10^{-5} and 10^{-3} changes the compliance by at most 0.053% and the sensitivities by at most 0.12% (penalty energy 1.2×10^{-4} to 5.6×10^{-4} of the total at $\gamma = 10^{-3}$), and refining the volume integration changes both by at most 0.010%. The central-difference stiffness derivative agrees with differences of the re-solved compliance to 4×10^{-8} , and relative to the step $10^{-5}\tau_c$ it changes by at most 2.6×10^{-7} at $10^{-3}\tau_c$ and 1.6×10^{-10} at $10^{-6}\tau_c$, with second-order behaviour and no change of integration branch at the fixed active set.

To verify the operator: the accuracy runs evaluate the network in single precision and the stiffness actions, smoothing and coarse solve in double precision (Table 1). On U2, M1, M2, H1 and H2, the bilinear form $q_i^T \hat{S} q_j$ is symmetric to a relative 9×10^{-9} , the returned work $q^T \hat{S} q$ agrees with the recovered-field energy $q^T F^T K F q$ to 5×10^{-9} , the deployed operator reproduces the training-time field to 1.1×10^{-7} , and the energy of normalised rigid modes is at most 2×10^{-11} of a typical deformation energy. The contraction of Section 5.1 requires $b \geq \lambda_{\max}(D^{-1}A)$: the power-iteration endpoint used by all GPU runs exceeds the converged largest eigenvalue by 2.1–5.0% on

all 80 validation geometries (Appendix D), whereas the Gershgorin bound exceeds it by factors of 11 to 20. Four of the per-cell margins in Table ST04 come from a host evaluation whose seeded generator gives a different power-iteration start. The containment is thus verified per geometry, not guaranteed a priori; symmetry, positive semidefiniteness and $\hat{S} \succeq S$ do not depend on it.

6.3. Dependence on geometry and loading

Consistent tractions, which represent surface loads and neighbour tractions, are the principal loading class; the ghost penalty carries below 0.05% of their exact field energy in the five cells of Table ST04. Equal nodal forces also load weakly supported nodes of small cut elements, placing 49–81% of the exact field energy in the ghost penalty, and serve as a stress test of the stabilised problem.

NICE removes most of the geometric dependence of the uncorrected extension (Figure 5). Under consistent tractions, its mean directional energy excess is 0.016%, 0.079%, 0.091% and 0.109% across the uncut, lightly, moderately and heavily cut strata (0.074% overall, largest geometry mean 0.65%), against 1.03%, 5.33%, 7.82% and 11.2% for the Uncorrected continuation (6.33% overall, largest 42%) and 6.89% for the base network (largest 48.6%). Neighbour-induced displacements give 0.060% (largest 0.38%) and stiffness-scaled spring supports 0.057% (0.47%); for single-face nodal forces the largest geometry mean is 0.72% (Table ST03). Under nodal forces, the base network's mean rises from 0.908% in uncut to 11.3% in heavily cut cells (largest 70.1%), whereas NICE's stratum means stay between 0.0135% and 0.0897% (largest 0.325%). On the 60 geometries outside weight selection, NICE's overall means are essentially unchanged (0.077% under consistent tractions, 0.0591% under nodal forces), but its cut-stratum means are higher (0.105% moderate, 0.126% heavy) and the five largest geometry means all occur there (Supplementary Table ST03d). Individual sampled directions are not bounded by these geometry means: over the 5,120 sampled consistent-traction directions of the 80 geometries, NICE's directional energy excess has a 95th percentile of 0.33%, a 99th percentile of 0.69% and a maximum of 1.24%, and over the 3,840 directions of the 60 geometries outside weight selection 0.35%, 0.73% and 1.24%.

The intermediate predictors locate this improvement. Continuing the base network without correction changes its mean only from 6.89% to 6.33%, and Smoothing-trained reaches 1.28% (largest 7.8%). NICE-post

gives 0.0965% (largest 0.91%), so most of the improvement comes from the correction itself. NICE's mean is lower than NICE-post's by a factor of 1.31 (95% bootstrap interval over geometries 1.20–1.40; one training seed), and by 1.5–2.0 under nodal forces, supports and single-face loads, on 96–99% of the geometries.

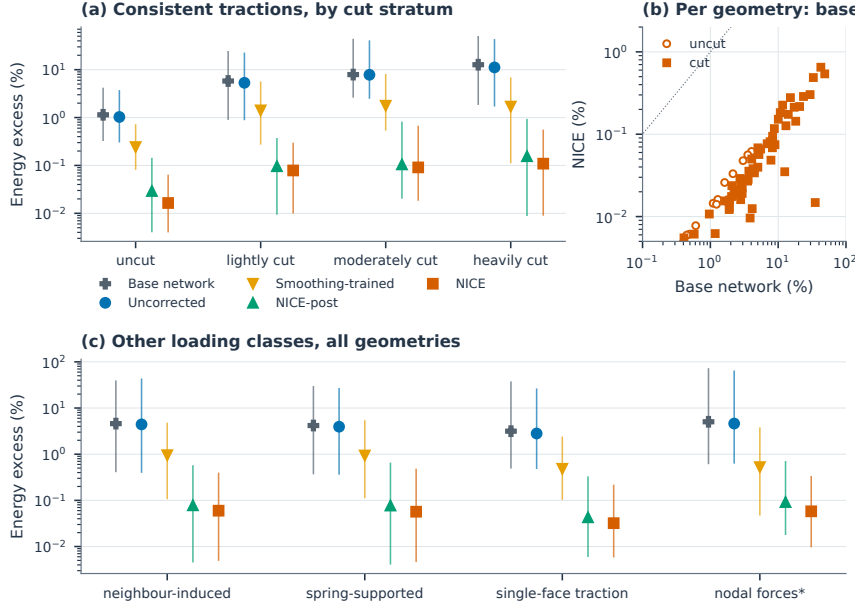


Figure 5: **Directional energy error of the learned substructures on the 80 validation geometries.** Each observation is a geometry's mean directional energy excess $q^T(\hat{S} - S)q/(q^T S q)$ over the validation directions of a loading class. (a) Consistent tractions, 20 geometries per cut stratum: markers give the mean over geometries and bars the range from the 10th percentile to the maximum. (b) Geometry means of the base network and of NICE under consistent tractions, open markers for uncut and filled markers for cut geometries; the dotted line denotes equality. (c) Neighbour-induced retained displacements, stiffness-scaled spring supports, single-face consistent tractions and equal nodal forces (75 geometries for the first two classes, 80 for the others); the asterisk marks the nodal-force stress test. Twenty geometries (6 uncut, 14 cut) entered weight selection. Predictors as in Table 2.

6.4. Components of the extension error

The base network's error occupies different parts of the Jacobi-scaled internal spectrum in different cells (Figure 6). Under consistent tractions on M1, the lowest 200 modes contain 24.5% of the error energy but only 4.87%

of the exact-field energy, and all lie below the lower endpoint $a = 0.173$ of the smoothing interval; in H2 only 66 of the 200 modes lie below $a = 0.138$. This is consistent with the smoothing rates of Section 6.5 (Table ST05b).

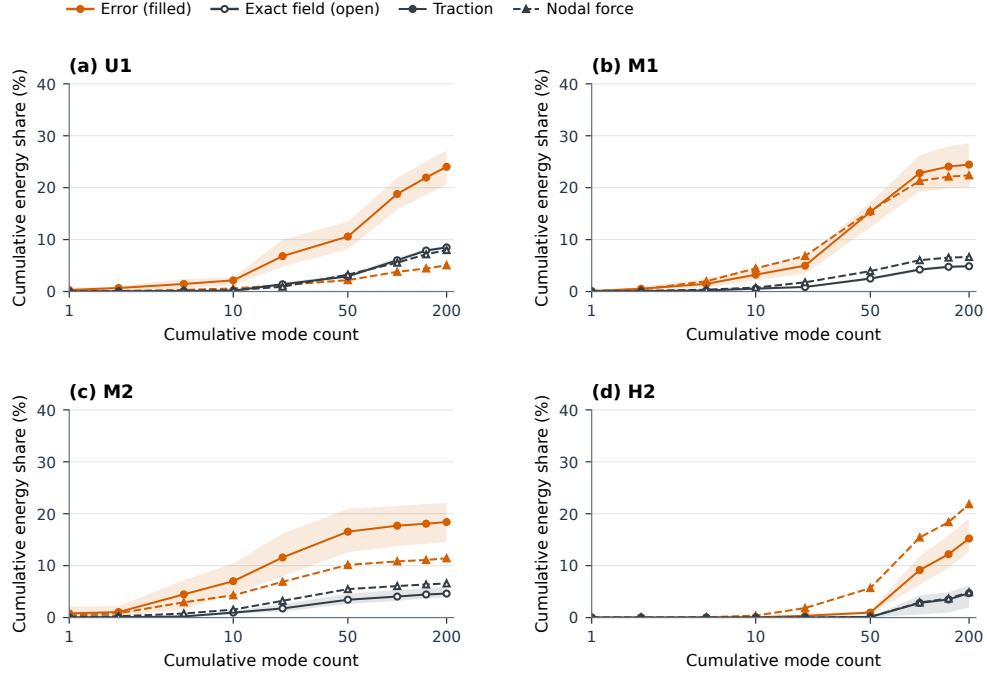


Figure 6: **Spectral distribution of the uncorrected extension error.** Panels show U1, M1, M2 and H2 for the base network. Modes solve $Av = \lambda Dv$, with $D = \text{diag}(A)$, ordered by increasing eigenvalue. Filled orange markers represent the extension error and open grey markers the exact internal field; solid circles correspond to consistent tractions and dashed triangles to nodal forces. Curves are directional means; bands give the 10th–90th directional percentiles under consistent tractions. Each cumulative fraction uses the total internal energy of its own field or error.

Spatially (Figure 7), the exact field of M1 under one consistent-traction direction places 8% of its energy in the elements within two element widths of the cut plane, which are 13% of the elements, whereas 39–43% of the base network's error energy lies there, next to the retained cut-band coordinates whose values the extension must propagate. The corrected field reduces the total error energy by about two orders of magnitude (113–138 times) and the element error energies by roughly 25 to 300 times (10th–90th percentiles), and halves the share of this layer to 19–22%.

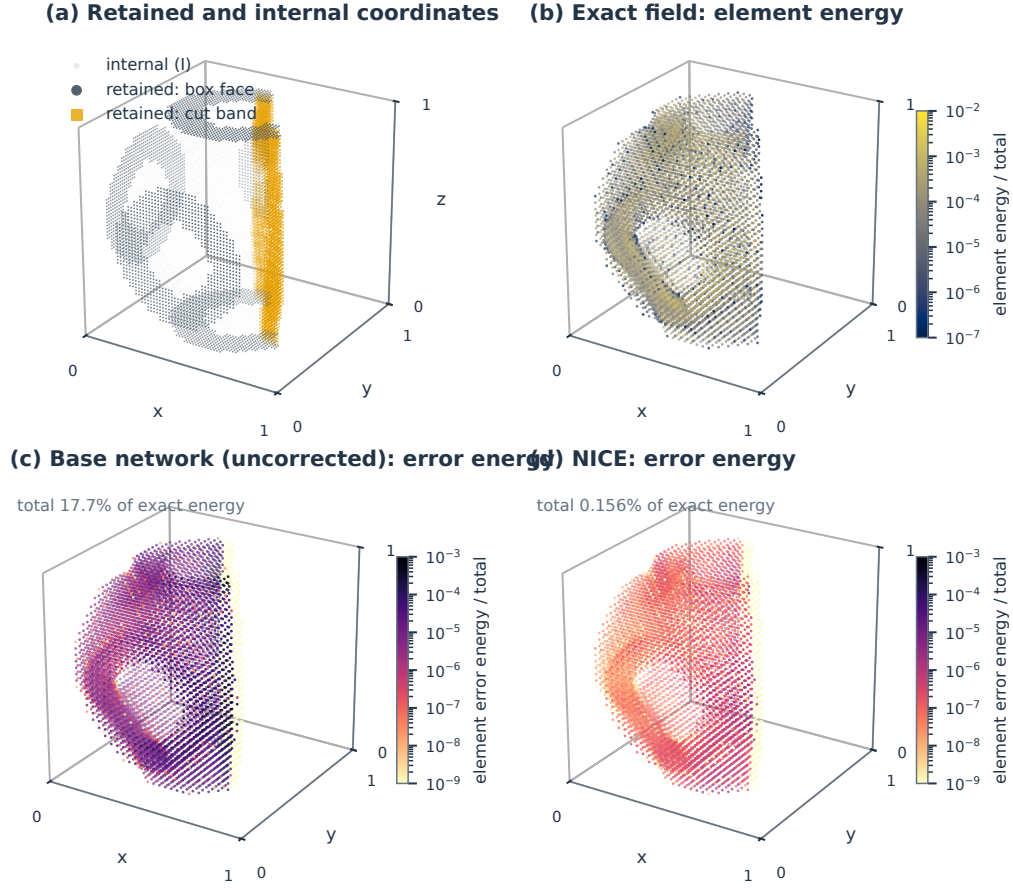


Figure 7: **Retained coordinates and spatial distribution of the extension error in M1.** (a) Retained box-face coordinates, retained cut-band coordinates and internal coordinates. (b) Bulk element energies of the exact field under one consistent-traction direction, relative to its total. (c,d) Bulk element energies of the error of the base network and of NICE for the same retained displacement, on a common colour scale. Cut-band elements carry no error because their coefficients are retained.

The size of these energy errors reflects a measurable amplification. For interior error d_I and exact interior field u_I , let $\delta^2 = d_I^T D d_I / u_I^T D u_I$ be the Jacobi-weighted relative displacement error and $\kappa = R(d)/R(u)$ the ratio of the Rayleigh quotients $R(d) = d_I^T A d_I / d_I^T D d_I$ and $R(u) = u^T K u / u_I^T D u_I$. Then $\varepsilon = \delta^2 \kappa$ exactly, with the weighting $W = \text{diag}(0, D)$ of Appendix B.1, admissible because d vanishes on the retained coordinates; the identity is a diagnostic decomposition, not an independent explanation. Under consistent tractions, the base network's fields in M1, M2, H1 and H2 have δ of 0.7–1.3% but κ of 124 to 7089: the displacement error is small but lies in directions far stiffer, relative to their amplitude, than the exact field, so that energy errors of 2–35% result. In the uncut U2 the same δ of 0.74% meets $\kappa = 85$ and gives 0.40%. NICE's fields have δ of 0.05–0.19% and κ of 33 to 582 in the cut cells (0.11% and 47 in U2).

At fixed retained displacement, M1 has mean energy and field-based sensitivity errors of 13.5% and 14.1%, and H2 of 35% and 75.1%; the linear term of Eq. (13) contributes only 7.2% and 0.79% of the sum of the linear- and quadratic-term norms, so the quadratic term dominates these finite errors (Table ST05, Figure S03).

6.5. Reducing interior error with fixed network weights

Holding the base network fixed isolates the effect of the correction (Figure 8). Eight Chebyshev steps at fixed retained displacement reduce the mean energy excess everywhere, but unevenly: under consistent tractions H2 falls from 35% to 0.205%, whereas M1 falls only from 13.5% to 4.68% (2.95% after 32 steps), consistent with its low-mode error below the smoothing interval; M1's sensitivity error falls from 14.1% to 2.76%, non-monotonically (Figure S04). A trilinear coarse correction before the same eight steps brings M1 to 0.230%, and eight further pre-smoothing steps to 0.186%, with 5,601 coarse coefficients for 165,927 internal degrees of freedom. Two, four and eight steps per stage give 1.02%, 0.422% and 0.186% on M1, and the eight-step cycle about 0.027% on M2 and U1; other coarse spaces are compared in Table ST07 and Supplementary Note S3.

Table 3 applies the same correction to four starting fields at the same retained displacements: a zero interior, a graph-harmonic extension (a volume-weighted graph Laplacian on the element connectivity, with the same exact rigid split), the base network's field (NICE-post) and NICE. A zero interior leaves 28–1100% energy excess and the harmonic start 0.08–17%, whereas the base network's field ends at 0.004–0.19%, lower by factors of 7 to 290

than the harmonic start and 2,400 to 12,000 than the zero interior. With 64 steps per stage, eight times the smoothing work, the harmonic start remains 2.6 to 50 times above the base network's eight-step result in five of six cells (M1, M2, H1, U1, U2) and reaches it only in the heavily cut H2; under nodal forces it ends above the zero interior at 16 and 32 steps in U2 and from 16 steps on in H2, so it is a weak but inexpensive comparator (Tables ST08 and ST08b). The learned field thus supplies the part of the interior equilibrium that smoothing and the coarse space do not reach at a practical budget. Locally, NICE-post is about as accurate as NICE.

Table 3: Mean energy excess (%) after the same 8 / $Q_1(17)$ / 8 correction applied to different starting fields (consistent tractions, 32 directions).

Cell	Base network, uncorrected	Zero interior	Harmonic	Base network (NICE-post)	NICE
M1	13.5	1097	17.2	0.186	0.131
M2	3.64	324	6.72	0.0273	0.0358
H1	1.81	39.9	1.28	0.0131	0.0115
H2	35	28.3	0.0806	0.0117	0.0148
U2	0.402	47.2	1.23	0.00424	0.00465

The first column gives the base network's error without correction. Results with 32 and 64 smoothing steps per stage and under nodal forces are given in Table ST08.

6.6. Compliance and local sensitivity after assembly

Figure 9 compares the predictors in the two-cell configurations of Figure 4. NICE meets both references in all fourteen configurations of the seven development cells: its largest compliance error over the six face loads is 0.056% (M1/x), and its largest sensitivity error, taken over both cells, is 0.67% (U1/y). On nine validation cells outside weight selection, chosen before evaluation as the five geometries with NICE's largest single-cell errors and one random cell in each of the uncut, lightly, moderately and heavily cut strata, the eighteen configurations give maxima of 0.27% in compliance and 1.49% in sensitivity, both on the geometry with the largest single-cell error (Supplementary Table ST09b). The five largest-error geometries account for all compliance errors above 0.02%; the four random cells stay below 0.011%

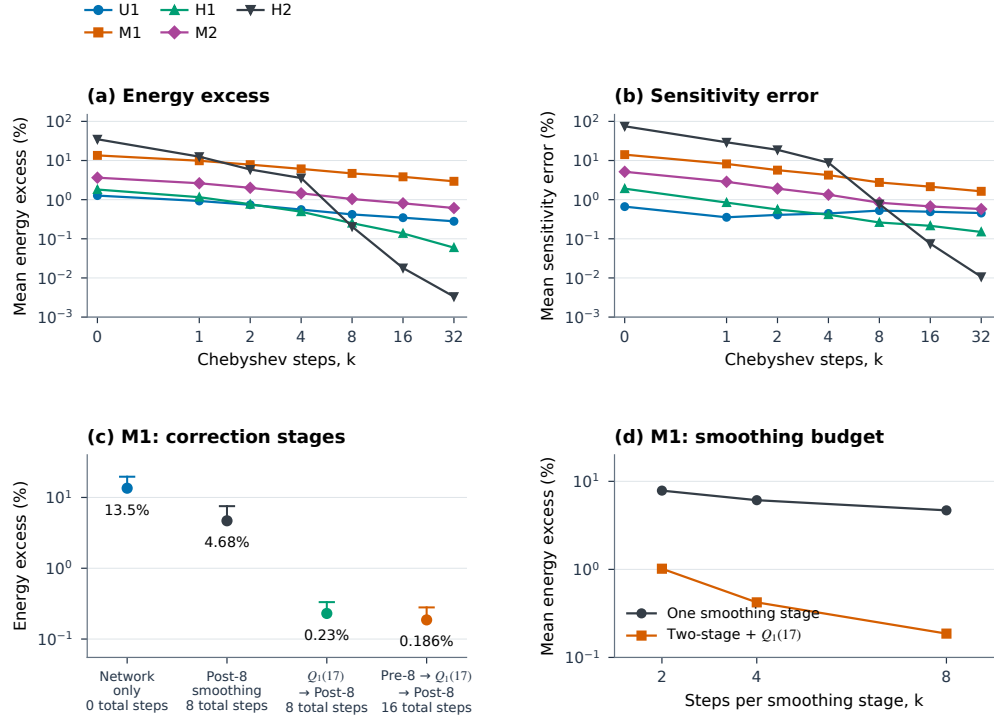


Figure 8: **Accuracy gained by correcting the base network at fixed weights.** (a,b) Mean directional energy excess and field-based sensitivity error during Chebyshev smoothing on five cells. (c) M1 with no correction, eight smoothing steps, coarse correction followed by eight steps, and eight steps on each side of the coarse correction; dots show means and caps the 90th percentile. (d) Mean energy excess versus steps per smoothing stage. All panels use consistent tractions, fixed retained displacements and $a = b/30$; in (a,b) the step axis is linear from zero to one and logarithmic thereafter.

and 0.22%. Assembled accuracy therefore follows the single-cell error, and remains within both references on every held-out configuration. The Uncorrected continuation was evaluated on eleven configurations, the nine common to Uncorrected, Smoothing-trained, NICE-post and NICE (the base network was evaluated on seven of them) and L1/x and L1/y; it fails the sensitivity reference in four of them (U1/x, U1/y, M1/x, M1/y, with up to 11.4%) and, on M1, also the compliance reference (up to 4.1%). Smoothing-trained reduces these errors but still fails the same four configurations (3.15–4.43%): smoothing without the coarse solve does not reach the slowly relaxed error components that dominate the sensitivity of U1 and M1. NICE-post also meets both references in the fourteen configurations (largest sensitivity error 0.95%, U1/x), so the correction is what brings the assembled responses within the reference. The largest sensitivity errors of NICE and NICE-post are 0.16–0.21% and 0.34–0.35% on M1, 0.60–0.67% and 0.90–0.95% on U1, and 0.136–0.154% and 0.075–0.083% on M2; NICE has the lower maximum in ten of the fourteen configurations (Table ST09). Under the cut-face tractions, NICE reaches at most 0.10% in compliance (M1/y) and 0.16% in sensitivity, whereas the Uncorrected continuation exceeds the reference on L1/y with 3.7% (Table ST10).

Table 4 shows why both quantities are needed. Without the complete correction, an accurate compliance does not guarantee an accurate local sensitivity: under a neighbour-face load on U1/x, the Smoothing-trained compliance error is 0.00109% while the target-cell sensitivity error is 3.15%, with the target carrying 0.13% of the exact assembled energy. NICE reduces both errors: the target-cell sensitivity error of that load falls to 0.598%, and on M1/x under the target-face y load NICE gives 0.0481% and 0.145% instead of the Uncorrected 3.44% and 11.4%. The separation remains visible in NICE, whose sensitivity error exceeds its compliance error by a factor of about two on H1 and by four orders of magnitude for the weakly participating U1 target, but both lie well within the reference.

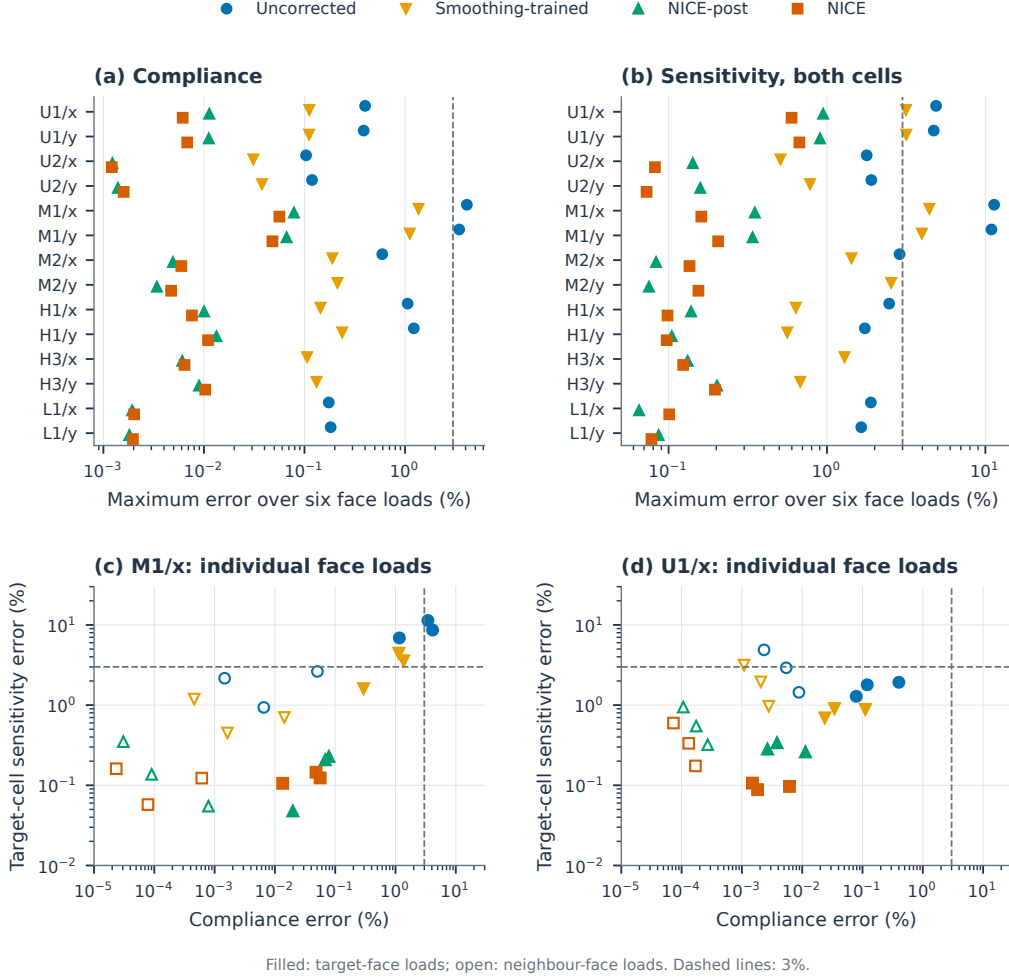


Figure 9: **Compliance and thickness sensitivity in assembled cell pairs.** The target cell uses a learned operator and the neighbour exact condensation. (a,b) Maximum errors over the six face loads in each configuration; the sensitivity error is also maximised over both cells. (c,d) Compliance and target-cell sensitivity errors of the individual face loads on M1/x and U1/x; filled markers denote target-face loads and open markers neighbour-face loads. Dashed lines mark the 3% reference. H3 is a further heavily cut cell evaluated for Smoothing-trained, NICE-post and NICE only; L1 is a lightly cut cell evaluated for Uncorrected, NICE-post and NICE. Predictors as in Table 2.

Table 4: Compliance and target-cell sensitivity under individual face loads.

Target / configura- tion	Load	Uncorrected: compliance / sensitivity (%)	Smoothing- trained: compliance / sensitivity (%)	NICE: compliance / sensitivity (%)	Target energy share
H1/x	T-x	1.06 / 2.47	0.144 / 0.636	0.0076 / 0.0133	0.643
U1/x	N-z	0.0023 / 4.89	0.00109 / 3.15	7.3×10^{-5} / 0.598	0.0013
M1/x	T-y	3.44 / 11.4	1.14 / 4.43	0.0481 / 0.145	0.311

T and N identify the loaded face of the target or neighbour; x, y and z give the traction direction. The last column gives the target cell's share of the exact assembled energy. Configuration-level sensitivity maxima include both cells.

6.7. Energy participation

Energy participation explains how a local stiffness error can have little effect on compliance. Under the neighbour-face z load in U1/x, the base network's target cell carries 0.127% of the exact assembled energy and has a local energy excess of 2.31%; their product $\beta = w\varepsilon$ gives a relative compliance bound of 0.00295%, close to the observed 0.00283%, whereas the target-cell sensitivity error is 5.83%. Figure 10 shows this weighting across predictors and loads. For the sensitivity at fixed retained displacement, Appendix J.4 gives $\|\tilde{\mathbf{s}} - \mathbf{s}\|_2 \leq 2L\sqrt{\mathcal{E}} + Q\mathcal{E}$, with $\mathcal{E}$ the interior error energy, L the coupling of the error to the stiffness derivative acting on the exact field and Q the derivative relative to the interior stiffness; the relative error is further scaled by the local reference $\|\mathbf{s}\|_2$, which is small for a weakly participating cell. In assembly the changed retained displacement also interacts with the extension error, so field-only and solution-only replacements do not add to the full sensitivity error (Table ST11, Appendix H).

6.8. Ablation: restricting the retained box-face representation

To isolate the role of the complete retained space, pairs in configuration x are solved with exact cell operators while the box-face displacements of each

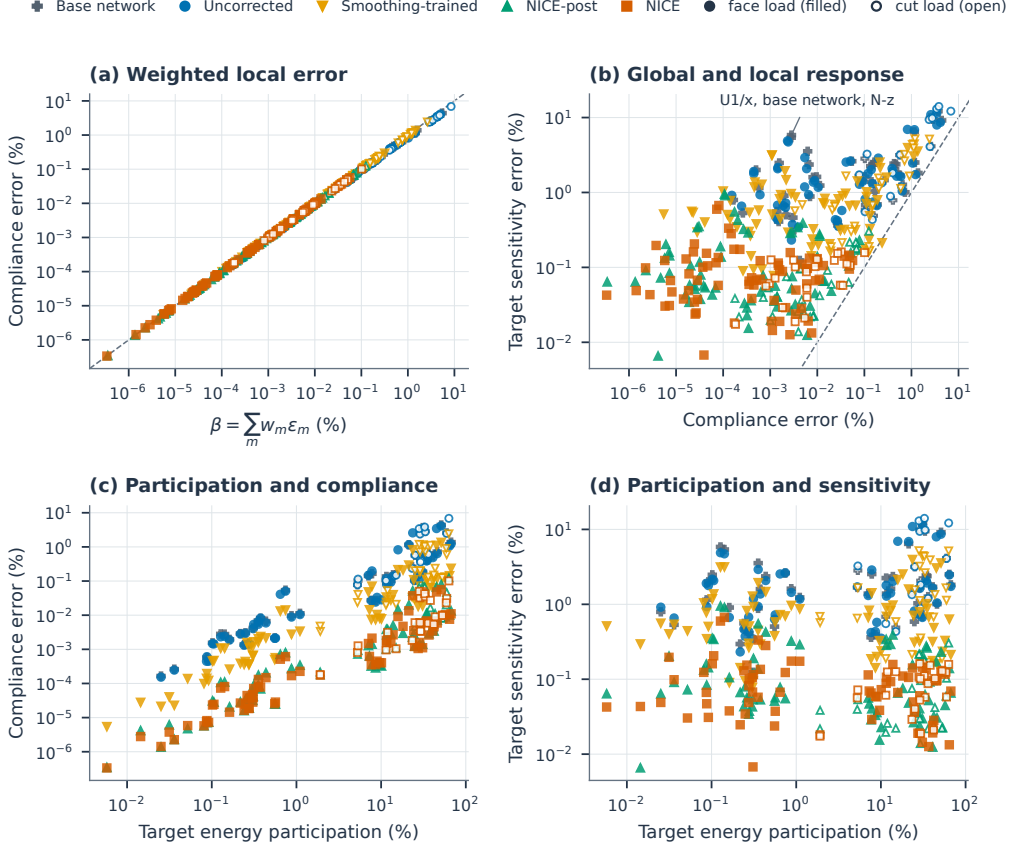


Figure 10: **Participation-weighted compliance error and local sensitivity.** Each point is one load for one predictor–configuration combination: 411 observations from 52 combinations of the base network, Uncorrected, Smoothing-trained, NICE-post and NICE in Table ST09, excluding L1. Filled markers denote face loads and open markers cut-surface loads. (a) Compliance error against $\beta = \sum_m w_m \varepsilon_m$, with $w_m = q_m^T S_m q_m / C$ and $\varepsilon_m = q_m^T (\hat{S}_m - S_m) q_m / (q_m^T S_m q_m)$, evaluated at the exact assembled retained displacement; only the learned target contributes to β . (b) Compliance and target-cell sensitivity errors under the same loads; the annotation identifies the base network on U1/x under the neighbour-z load. Dashed lines denote equality. (c,d) The two response errors versus the target's exact energy participation.

cell are restricted to tensor Bernstein polynomials of degree r ; the non-box cut-band coordinates remain unrestricted. This ablates our own retained representation at a fixed substructure size of one cell, under the fine-scale face tractions used throughout and without oversampling, and compares accuracy, not accuracy at equal cost; it is not a model of reduced-boundary substructure methods, which control this error by partition refinement, boundary enrichment or oversampling Huang et al. (2024), Guo et al. (2026a), Guo et al. (2026b) (Supplementary Note S4). With every box face restricted, degree one gives compliance errors of 78–85% on U1, M1, M2 and H1, decreasing to about 4% at $r = 5$ and 0.49–0.74% at $r = 8$, where the H1 pair retains 14,001 of 32,991 coordinates (Figure 11). The local sensitivity converges more slowly: at $r = 8$ the target-cell sensitivity error is 1.3–4.5% under target-face loads and 24–64% when neighbour-face loads are included. Restricting only the shared interface removes most of the compliance error (0.17–0.44% at $r = 3$ on U1, M1 and M2), but the sensitivity error under neighbour loads remains 8–21% at $r = 3$ and 2.2–3.4% at $r = 5$. A restricted interface is thus adequate for compliance at low degree but not for local sensitivity under neighbour loads; keeping the complete retained space removes this component of the error, at the cost of the larger coarse model (Table ST16).

6.9. Heterogeneous lattices with every cell learned

In a design every cell is learned, and the errors of neighbouring cells enter the same assembled solution. Two lattices with a continuous graded thickness field and a planar boundary cut test this: a $2 \times 2 \times 2$ block of eight distinct cells, four of them cut with retained volumes of 62% and 25%, and a $3 \times 3 \times 1$ layer of eight cells, three of them cut, with one corner position left empty. Neighbouring cells share their face corners, the corner parameters of the block range from 0.25 to 0.54, and the sixteen cells were generated for these examples and entered neither training nor weight selection. The lattices are clamped on one face and loaded by unit consistent tractions on the opposite face, held fixed at the evaluated design for the sensitivities (Appendix H), and by three random load vectors. Every cell uses NICE; the reference assembles the dense exact condensed matrices and solves to a recomputed relative residual of at most 1.3×10^{-10} . The learned lattices are solved by preconditioned conjugate gradients on the free retained coordinates with a balanced two-level preconditioner: the fine action is the inverse of the assembled retained stiffness block K_{PP} , factorised once per design iteration,

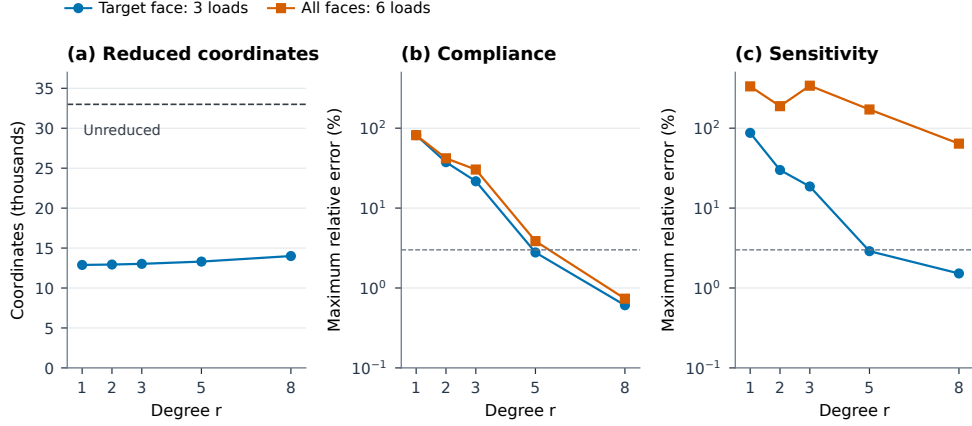


Figure 11: **Response errors caused by restricting box-face displacements.** Both cells of H1/x use exact operators and Bernstein degree r on every box face, with unrestricted non-box cut-band coordinates. (a) Reduced coordinate count; the dashed line denotes the 32,991-coordinate full representation. (b,c) Maximum compliance and target-cell sensitivity errors over the three target-face loads or all six target- and neighbour-face loads; macro-cut tractions are excluded. Errors are relative to the full retained-space solution; horizontal lines mark 3%.

and the coarse space consists of trilinear macro-vertex functions multiplied by the six rigid-body modes (Supplementary Note S6.1).

In the $2 \times 2 \times 2$ block, the lattice compliance errors under the three face loads are 0.014%, 0.011% and 0.0094% (at most 0.069% under the random loads), the largest eight-corner sensitivity error over all cells is 0.14% (0.45% under the random loads), and the assembled retained solution differs from the exact one by 0.13%. The $3 \times 3 \times 1$ layer gives 0.015%, 0.013% and 0.010% (at most 0.056%), 0.14% (0.43%) and 0.12%. These errors are below the largest two-cell errors: the cells' energy errors combine through their shares of the assembled energy rather than accumulate. By Eq. (12), the lattice compliance error is bounded by the participation-weighted sum of the cells' energy errors at the exact traces, which range from 0.006% to 0.026% in the block and 0.005% to 0.052% in the layer under the face loads (0.02% to 0.12% under the random loads), largest in the cut cells. Their weighted sums exceed the lattice compliance errors by less than 0.4% of their value under the face loads and by about 4–5% under the random loads, because the assembled solution relaxes towards the stiffer surrogate whereas β evaluates the errors at the exact traces.

The learned solves reach the prescribed recursive residual, while the re-computed residual $\|f_g - \widehat{\mathbb{K}}\bar{U}\|/\|f_g\|$, at which the errors above are measured, stagnates at 3.6×10^{-4} (block) and 3.2×10^{-3} (layer). Repeating the block solve with the correction in single precision changes this residual by less than 1% and the compliance errors by less than 3×10^{-9} , so the stagnation does not come from the correction arithmetic; it is consistent with rounding in the single-precision network. The solve-independent bound β exceeds the errors by only $1.5\text{--}4.9 \times 10^{-7}$ of the compliance on all face loads, which leaves no room for a residual work comparable to them. Instrumented re-solves confirm this directly (Supplementary Note S6.3, Table ST19): the signed residual work $\bar{U}^T \rho$ of Eq. (18) is at most 2.5×10^{-8} of the compliance, its dual-norm bound at most 1.4×10^{-5} , and the terms of Eq. (18) balance to 9×10^{-9} of the compliance, with the compliance error unchanged from a recursive residual of 10^{-3} onwards. Aggregated over the shared corner parameters, the field-based sensitivities give the lattice gradient to 0.04–0.09% under the face loads and 0.22–0.28% under the random loads, with cosines above 0.999999.

6.10. Computational cost of a lattice design iteration

Cost is compared for one design iteration of the lattices of Section 6.9: the two eight-cell lattices and, as four-cell lattices, the two $2 \times 2 \times 1$ layers of the block, each with two uncut cells and two cut cells of 62% and 25% retained volume, with the same retained coordinates, clamp and loads. Table 5 compares three routes. (a) The whole-lattice direct solution assembles the full cut finite element model of the lattice, retained and interior degrees of freedom of every cell, and factorises it once by a sparse Cholesky factorisation (MKL PARDISO on one AMD EPYC 9654 host with 16 threads, and with one thread for comparison with serial solvers); its recomputed relative residuals are below 3×10^{-9} , so for the four-cell lattices it is also the exact reference. (b) Conventional exact condensation condenses every cell on the same host by a Cholesky factorisation of its interior with PARDISO's Schur-complement option and solves the assembled condensed system by a block Cholesky factorisation that eliminates each cell's private retained coordinates before the shared interface (16 threads); its compliance agrees with (a) to 7×10^{-10} . (c) NICE (one RTX 5090) prepares the learned operator of every cell and solves the assembled system by the preconditioned conjugate gradients of Section 6.9 to a relative residual of 10^{-6} , with the correction in single precision, including the field-based sensitivities. All routes include

cell setup and stiffness assembly, and none includes the one-off cost of data generation and training (Section 6.1).

Table 5: Cost of one design iteration of the lattices. (a) Whole-lattice direct solution by sparse Cholesky factorisation (MKL PARDISO, AMD EPYC 9654) with 16 threads and with one thread: time from cell setup to the solution of six loads (three consistent, three random); peak process memory. (b) Every cell condensed with PARDISO's Schur-complement option, then the condensed lattice system solved by block Cholesky factorisation, 16 threads: total time, peak process memory. (c) Front end, preconditioner setup, conjugate gradients for the three consistent loads to 10^{-6} and field-based sensitivities; peak GPU memory, host memory after the front end; compliance error against (a) for four cells and against the exact condensation of Section 6.9 for eight cells. Memory in GiB. All phases: Supplementary Note S5 and Table ST17.

Lattice (cut / cells)	DOFs: total / free retained	(a) Whole-lattice direct, 16 threads: time (s) / memory (GiB)	(a) Whole-lattice direct, 1 thread: time (s)	(b) Conventional exact condensation, 16 threads: time (s) / memory (GiB)	(c) NICE (one RTX 5090): time (s) / iterations	(c) NICE memory (GiB): GPU / host	(c) NICE compliance error (%)
$2 \times 2 \times 1$, $z = 0$ (2 / 4)	957,888 / 77,310	484 / 35.1	2,425	589 / 27.9	38.6 / 114	6.3 / 2.4	0.015, 0.012, 0.010
$2 \times 2 \times 1$, $z = 1$ (2 / 4)	884,940 / 71,046	349 / 32.0	2,131	632 / 24.7	37.7 / 119	6.1 / 2.4	0.022, 0.016, 0.014
$2 \times 2 \times 2$ (4 / 8)	1,833,474 / 139,002	868 / 71.1	5,864	950 / 40.1	81.2 / 129	8.5 / 3.9	0.014, 0.011, 0.0097
$3 \times 3 \times 1$ (3 / 8)	2,113,611 / 143,685	1,042 / 85.9	8,131	1,218 / 42.2	110.1 / 165	9.2 / 4.7	0.015, 0.014, 0.015

For the four-cell lattices, a NICE design iteration takes 38 s, against 349 to 484 s for the whole-lattice direct solution with 16 threads, 2,131 to 2,425 s with one thread and 589 to 632 s for conventional exact condensation, and the learned compliance agrees with the direct solution within 0.022%, below the reference in every case. For the eight-cell lattices, NICE takes 81 and 110 s against 868 and 1,042 s for the parallel and 5,864 and 8,131 s for the serial direct solution, about ten and 72 to 74 times longer, and 950 and 1,218 s for conventional condensation. For the eight-cell lattices, the timed errors include the algebraic error of the 10^{-6} solve (Eq. 18) and exceed the operator errors of Section 6.9 by up to 0.005 percentage points. The Cholesky factor grows slightly faster than the number of degrees of freedom, from 29 to 32 GiB for four cells to 66 and 81 GiB for eight, and the process peak of the direct solution reaches 71 and 86 GiB; conventional condensation needs 40 and 42 GiB, most of it for the condensed cell matrices held until assembly (Supplementary Table ST17). NICE needs at most 9.2 GiB of GPU and 4.7 GiB of host memory. Per cell, NICE's stored state takes 0.25 to 1.37 GiB, against 0.60 to 9.4 GiB for the interior Cholesky factor of conventional condensation (Supplementary Note S5, Table ST18). SciPy's default sparse direct solver (SuperLU), tried as a further serial baseline, failed with a memory error while factorising a single cell, at about 4 GiB of resident memory.

A design iteration changes the geometry of every cell, so every cell's preparation, interior factorisation or network encoding and correction setup, is repeated; the assembled solve can start from the previous solution, and the conventional routes can reuse their symbolic factorisation while the active-element pattern is unchanged. NICE's lattice solve is itself an inexact substructuring iteration; fine-scale iterative solvers of the whole lattice were not compared (Section 7.5).

6.11. Design optimisation [placeholder]

[Placeholder — to be completed; not part of the current review.] This subsection will report a thickness optimisation driven by the learned operator: compliance minimisation under a volume constraint, with the eight corner thickness parameters of each cell as design variables shared between neighbouring cells, updated by optimality criteria or the method of moving asymptotes, on a lattice of eight or twenty-seven cells. For shared design variables $\boldsymbol{\tau}_g$ enforcing continuity through $\boldsymbol{\tau}_m = \boldsymbol{\tau}_m(\boldsymbol{\tau}_g)$, the structural gradient is $\nabla_{\boldsymbol{\tau}_g} C = \sum_m (\partial \boldsymbol{\tau}_m / \partial \boldsymbol{\tau}_g)^T \mathbf{s}_m$ under the fixed load and assembly

maps of Section 4.3; the same parameter map acts on the complete derivatives in Eq. (14). The final design will be verified with the exact reference (compliance and thickness sensitivities), and the optimisation history will be compared with one driven by exact condensation.

7. Discussion

7.1. Boundary restriction and interior approximation

Fixing the retained coordinates separates two decisions that a reduced substructure model otherwise combines: which displacement patterns neighbouring cells may exchange, and how accurately the interior responds to each of them. Learning and correction alter only the second, which matters for cut cells, where the coefficients needed for cut-surface virtual work form a band of active elements rather than a minimal surface trace. The ablation of Section 6.8 shows the role of the first: even with exact interiors, restricting box-face displacements changes the assembled response, and a degree that gives accurate compliance can leave appreciable sensitivity error. Substructure methods that describe the boundary by a few coordinates Huang et al. (2023), Guo et al. (2026a) accept this boundary error, which they control by partition refinement Huang et al. (2024), boundary enrichment Guo et al. (2026a) or oversampling Guo et al. (2026b), in exchange for coarse models small enough for very large structures. For orientation only, since error definitions, problems and baselines differ: Guo et al. (2026a, Table 1) report displacement errors of 7.76–13.64% with linear, 1.95–2.22% with cubic Bézier and 0.84–3.46% with full-node boundaries, at 7 to 380 times the speed of their serial direct finite element solution, whereas NICE's directional energy error is about 0.07% and its assembled compliance errors are 0.01–0.06% against exact condensation, at 9 to 13 times the speed of the parallel and 56 to 74 times that of the serial whole-lattice direct solution (Section 6.10). For the local sensitivity of the cut cells examined, the boundary restriction produces larger errors than the interior approximation of NICE (Sections 6.6 and 6.8, for single cells at fixed size); where a compact coarse model matters more, the balance can differ. The retained cut band keeps cut-surface loads and supports available without retraining; in the lattices, whose cut surfaces are traction-free, it accounts for 37 to 50% of the free retained coordinates, but each cell application acts on the complete cell irrespective of how its retained coordinates are shared.

7.2. Complementary roles of prediction and equilibrium correction

An internal update that does not increase the A -energy error for any retained input yields a condensed stiffness between the learned stiffness and the exact Schur complement, independently of the network. The guarantee is an ordering, not a rate: the reductions by one to two orders of magnitude in Section 6.5 are empirical, and the ordering imposes no monotonicity on the sensitivity error. Within this structure, prediction and correction are complementary. Relaxation efficiently removes the error components with large Rayleigh quotients, which amplify a displacement error of one percent into an energy error of several percent (Section 6.4); the coarse solve removes the smooth components that relaxation reaches slowly. The learned field supplies what neither reaches at a practical budget: under the same correction, a harmonic start leaves 7 to 290 times its error, and eight times the smoothing work closes the gap in only one of six cells. Conversely, the correction relieves the network of what it represents poorly: a base-network error of 13.5% on M1 becomes 0.19% without any change of weights, and NICE-post already meets the assembled accuracy reference, so an existing network need not be retrained to benefit from the correction.

7.3. Requirements for design use

Design differentiation adds a requirement that the energy error does not control. At a fixed retained displacement, with $H = F - E$, Eqs. (13) and (14) give $\tilde{s}_c - s_c = 2(Hq)^T A E_{I,c} q - (Hq)^T K_{,c} H q$ and $\hat{C}_{,c} - C_{,c} = -2(H_{I,c} q)^T A H q - (Hq)^T K_{,c} H q$: the residual term of Eq. (14) removes the linear field-error term and replaces it by one weighted by the design derivative of the extension error. Which of the two is the more accurate gradient therefore depends on whether $\|H_{I,c} q\|_A$ is small compared with $\|E_{I,c} q\|_A$. A fixed- $\hat{q}$ central-difference check on the fourteen pair configurations and on the four lattices of Sections 6.9 and 6.10 could not resolve $\hat{C}_{,c}$ (Supplementary Note S6.3, Table ST19). Each perturbation rebuilds the cell's learned operator, and 21 to 48 of the 49 rebuilds of every lattice cell change at least one of the discrete choices listed in Section 4.3; the difference quotient then grows as the step decreases, from 0.18–0.29% of the exact lattice gradient at a step of $3 \times 10^{-3} \tau_c$ to 0.35–1.5% at $3 \times 10^{-4} \tau_c$. At this resolution the surrogate objective is not smooth in the thickness. The field-based estimate needs no difference quotient and gives the shared-variable lattice gradient to 0.02–0.13% under the face loads and 0.22–0.35% under the random loads.

The field-based estimate used here therefore estimates the exact sensitivity and is not the gradient of the surrogate objective, an inconsistency that matters for optimisers with a line search. Moreover, assembly selects its directions through the coupled solution, whereas training controls the average error over sampled directions; a uniform operator bound would require quantitative coverage of the nonrigid retained space (Appendix J.7).

7.4. Computational value

A correction budget adequate for compliance can leave local sensitivity inaccurate, so the relevant cost is the total work needed to reach the prescribed accuracy of both. For the lattices examined, a NICE design iteration takes less time and memory than the whole-lattice direct solution and conventional exact condensation (Section 6.10). Its lattice solve is an inexact substructuring iteration, with approximate interior extensions as in inexact BDDC and FETI-DP methods Li & Widlund (2007), Klawonn & Rheinbach (2007). Its distinguishing output is a condensed operator per cell with a quantified and reducible error, not a faster fine-scale solve, and no advantage over iterative solvers of the whole lattice is claimed. Fusing the operations of an application or reducing the smoothing budget would lower its cost, at an accuracy cost that Section 6.5 quantifies.

7.5. Limitations

The trained operator applies to the setting in which it was trained and tested: Schwarz-P-type cells defined by Eq. (1), eight trilinear corner parameters in $[0.175, 0.699]$ with corner span and gradient norm at most 0.47 (Table ST02), at most one planar cut whose normal is a cube-symmetry image of $(\cos \vartheta, \sin \vartheta, 0)$, and one discretisation ($n = 32$, $E_Y = 1$, $\nu = 0.3$, $\gamma = 10^{-4}$), to which the 65/33/17/9 latent hierarchy is tied. Other TPMS families were not tested; they, cells with several cuts or curved boundaries, other resolutions and other materials require new training. The network is not equivariant under the cube symmetries, as symmetry-decoupled shape functions are for voxel substructures Jiang, C. et al. (2026), and NICE was evaluated mainly in the identity orientation: in a second cube-symmetry orientation of the 80 validation geometries, which also entered weight selection, its mean directional energy excess over the load classes is 4% higher, and 12% higher under consistent tractions. The variational properties of Section 3.2 hold for any input, but the accuracy does not: no certified error bound is provided, although the residual of Eq. (10) and a label-free lower bound

on each cell's energy error (Appendix I) can be computed at run time. Accuracy is measured against the discrete reference, whose error for heavily cut cells at $n = 32$ is of order one percent in compliance and several percent in sensitivity (Section 6.2). The assembled evidence comprises fourteen two-cell configurations of development cells, eighteen of held-out cells and two eight-cell lattices; all training comparisons use one seed, and the effect of the sensitivity term in Eq. (8) was not isolated. The reported sensitivities are field-based estimates that omit the design dependence of the extension in Eq. (14). The cost comparison excludes data generation and training, and fine-scale iterative solvers of the whole lattice, such as algebraic multigrid Vaněk et al. (1996), Henson & Yang (2002), Falgout & Yang (2002) or BDDC and FETI-DP preconditioners Dohrmann (2003), Farhat et al. (2001), were not compared. An optimiser must keep designs inside the parameter domain above.

8. Conclusions

NICE approximates the static condensation of cut thin-walled Schwarz-P-type TPMS cells on the complete retained space by a geometry-conditioned learned interior field and a fixed, geometry-specific multilevel equilibrium correction, applied through the complete transpose without forming the condensed stiffness. The operator inherits the variational properties of admissible extensions, including the Schur-complement lower bound, and the correction cannot increase its error; in all examples, a better initial field or a larger correction budget reduced it.

For learned substructures, the error relations quantify a distinction familiar from goal-oriented error estimation and approximate reanalysis: accurate compliance does not imply accurate local thickness sensitivity. For an uncorrected extension they also explain why: displacement errors of about one percent lie in stiff directions and give energy errors of several percent, and weakly participating cells show accurate compliance but sensitivity errors above 3%. Chebyshev smoothing and a trilinear coarse solve remove the stiff and the slowly relaxed error components while preserving the variational structure. NICE keeps the geometry-mean energy error under consistent tractions at or below 0.65% on all 80 validation geometries, the largest occurring on a geometry that entered neither training nor weight selection, and keeps the face-load compliance and sensitivity errors below 0.06% and 0.7%

in the fourteen two-cell configurations of development cells and below 0.28% and 1.5% in eighteen configurations of cells outside weight selection.

The learned field and the correction are complementary: under the same correction, a harmonic start leaves 7 to 290 times and a zero interior 2,400 to 12,000 times the error of the learned field on five detailed cells, and NICE-post, the base network with the correction applied at deployment, also meets the assembled accuracy reference. In lattices of learned cells, the compliance error, at most 0.015% under consistent face loads, follows the participation-weighted bound of Eq. (12) to within 0.4% under face loads (about 5% under random loads), and the thickness sensitivities stay within 0.15%. For the eight-cell lattices, a NICE design iteration takes 81 to 110 s and at most 9.2 GiB of GPU memory on one RTX 5090, against 868 to 1,042 s for the whole-lattice Cholesky solution on 16 CPU threads, whose factor needs 66 to 81 GiB, and 5,864 to 8,131 s on one thread; these figures exclude the one-off cost of data generation and training.

[Placeholder: conclusion on the design-optimisation example of Section 6.11.]

9. Code and data availability

The implementation of the neural extension, the equilibrium correction and its transpose, the cut-cell preprocessing and reference condensation, and the assembly, benchmark and figure scripts will be archived as a cleaned, frozen snapshot, together with the trained weights of the predictors of Table 2, the parameter files of all validation and lattice geometries and the result records from which every table and figure is generated, at [Zenodo DOI] upon acceptance. Training data are available from the corresponding author on reasonable request.

CRedit authorship contribution statement

[Author contributions (CRedit taxonomy).]

Declaration of competing interest

[Competing-interest statement.]

Data availability

[Data and code availability statement.]

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Appendix A. Discrete construction and metric definitions

Appendix A.1. Active coefficients and element integration

The active background elements use tensor-product Q_2 displacements, with 27 nodes and 81 displacement degrees of freedom per element. Active coefficients are stored in a fixed node-major x, y, z order. The retained box set is the union of the nine face nodes of every certified positive-area material patch. The cut set is the union of all 27 nodes of each active element carrying a positive-area macro-cut patch. Their union is deduplicated in active-node order. The off-plane cut-band coefficients are retained because their basis functions determine displacement and virtual work on the cut plane.

The discrete stiffness is assembled as

$$K(\eta) = \sum_e L_e^T K_e(\eta) L_e + \gamma \sum_{f \in \mathcal{F}_g} L_f^T G_f^T G_f L_f, \quad K_e(\eta) = \sum_{\alpha \in \{0, \dots, 4\}^3} M_{e\alpha}(\eta) T_\alpha. \quad (\text{A.1})$$

Here L_e, L_f extract element and ghost-face coefficients. The fixed elasticity templates $T_\alpha \in \mathbb{R}^{81 \times 81}$ use the physical basis derivatives and isotropic Lamé constants $\lambda_L = E_Y \nu / [(1 + \nu)(1 - 2\nu)]$ and $\mu_L = E_Y / [2(1 + \nu)]$. The 125 moments $M_{e\alpha}$ integrate local monomials with coordinatewise exponents from zero to four over the material part of an element. The ghost-face set, its factors and coefficient specify the stabilisation contribution. The sum defines the stabilised discrete energy, including its ghost-penalty contribution.

The ghost-face set $\mathcal{F}_g$ consists of internal background faces whose two neighbouring elements are active and are not both certified as completely filled with material. These faces lie within each substructure. For background-element width $h = 1/n$, the unit-coefficient stabilisation bilinear form is

$$g_h(v, u) = (\lambda_0 + 2\mu_0) \sum_{f \in \mathcal{F}_g} \sum_{j=1}^2 h^{2j-1} \int_f [\partial_{n_f}^j v] \cdot [\partial_{n_f}^j u] \, dA.$$

The derivative uses a common positive coordinate normal on both neighbouring elements, and the jump is the value on the first element minus that on the second. Integration covers the complete background face. The fixed templates use $E_0 = 1$, $\nu_0 = 0.3$, and their Lamé constants λ_0, μ_0 ; these match the material values of the 80 validation geometries. Each face integral uses a tensor-product three-point Gauss rule in its two tangential coordinates. Stacking the square-root-weighted jump evaluations for both derivative orders and all three displacement components gives $G_f \in \mathbb{R}^{54 \times 135}$, acting on the 45 distinct nodes of the two-element patch. Thus the second term in Eq. (A.1) is the matrix representation of γg_h .

All 80 validation geometries use $n = 32$, normalised Young's modulus $E_Y = 1$, Poisson's ratio $\nu = 0.3$, and ghost-penalty coefficient $\gamma = 10^{-4}$, with length expressed relative to the unit reference box. This gives 65 Q2 node positions per axis; the network's three transfer levels use 33, 17 and 9 positions per axis. The moment evaluator begins with 4^3 subcells per element and refines partial subcells once, then clips Kuhn tetrahedra using the tetrahedral rule parameter four. Appendix F.4 gives the full integration sequence. Training populations and the historical deployment benchmark are identified separately.

Appendix A.2. Response metrics and aggregation

With $K_{,c} = \partial K / \partial \tau_c$ on the stated fixed-coordinate design interval, the metrics are

$$\begin{aligned} \varepsilon(q) &= \frac{\hat{u}^T K \hat{u}}{q^T S q} - 1, & C &= f_g^T U, & \hat{C} &= f_g^T \hat{U}, \\ e_C &= \left| \frac{\hat{C}}{C} - 1 \right|, & \tilde{s}_c(\hat{u}) &= -\hat{u}^T K_{,c} \hat{u}, \\ e_s &= \frac{\|\tilde{\mathbf{s}} - \mathbf{s}\|_2}{\|\mathbf{s}\|_2}, & \mathbf{s} &= \left(-u^T K_{,c} u \right)_{c=1}^8. \end{aligned} \tag{A.2}$$

The local same-retained-displacement comparison uses $u = Eq$. The assembled comparison uses the recovered fields at the respective global equilibria, whose retained displacements generally differ. All relative denominators

are nonzero. A geometry-level energy score averages over the evaluated directions in one class; population means then give equal weight to each available geometry. A population maximum of geometry means is distinct from a worst-direction operator error.

The pair-assembly comparison joins a learned target to an exact neighbour with matched thickness parameters on the common face. Its six face-load cases define the selected 3% joint compliance/sensitivity criterion, with sensitivity maxima taken over both cells. Cut-traction cases are reported separately (Table ST10). Table 2 and Tables ST03, ST01 and ST09 identify the populations and loading sets, and Supplementary Note S6 specifies the two-cell configurations, the pair solver and the preconditioner of the lattice solves.

Appendix B. Variational identity, rigid kernel, and directional norms

Let the selectors satisfy $J_P^T J_P + J_I^T J_I = I_{n_a}$, and assume the symmetric stiffness in Eq. (2) has $A \succ 0$. For any field v with $J_P v = q$, there is a unique $w \in \mathbb{R}^i$ such that $v = Eq + J_I^T w$. Since $J_I K E = 0$,

$$v^T K v = q^T S q + w^T A w. \quad (\text{B.1})$$

This proves the energy-minimising property and uniqueness of Eq . Applying the same expansion to a linear admissible extension $F = E + J_I^T H$ yields Eq. (9). Moreover, $r_I = J_I K F q = A H q$, proving both expressions in Eq. (10). These identities require trace admissibility and the specified symmetric stiffness; a contraction property of the approximation is a separate condition.

Suppose additionally that $\ker K = \text{range } R$, R_P has rank six, and $F R_P = R$. The exact field associated with $R_P a$ is Ra , because it has that retained trace and satisfies internal equilibrium. Thus $E R_P = R$ and $H R_P = 0$. Rigid reproduction gives $\hat{S} R_P = F^T K R = 0$. Conversely, if $q^T \hat{S} q = 0$, positive semidefiniteness implies $F q \in \ker K$. Write $F q = Ra$; selecting the retained entries gives $q = R_P a$. Hence

$$\ker S = \ker \hat{S} = \text{range } R_P. \quad (\text{B.2})$$

The retained rigid coefficient map and deformation projector are

$$C_R = (R_P^T R_P)^{-1} R_P^T, \quad \Pi_P = I_p - R_P C_R, \quad q_d = \Pi_P q. \quad (\text{B.3})$$

Rigid reproduction and symmetry give

$$\hat{S} R_P = 0, \quad R_P^T \hat{S} = 0. \quad (\text{B.4})$$

For the construction in Eq. (5), $C_R R_P = I_6$ and $\Pi_P R_P = 0$. A linear raw displacement map sends zero to zero, so $\hat{E} R_P = R$. Corrections driven by the internal residual leave this field unchanged. These observations establish the two-sided rigid annihilation in Eq. (B.4) without altering any stiffness eigenvalue.

Let $Z \in \mathbb{R}^{p \times (p-6)}$ have orthonormal columns spanning the retained rigid complement, and define $S_* = Z^T S Z \succ 0$. The all-direction energy error on this space is

$$\varepsilon_* = \sup_{q \perp R_P, q \neq 0} \varepsilon(q) = \|A^{1/2} H Z S_*^{-1/2}\|_2^2, \quad \mu_* = 1 + \varepsilon_*. \quad (\text{B.5})$$

Let $q = Z a$, with $Z^T Z = I$ spanning the rigid complement. Its relative energy excess is

$$\frac{a^T Z^T H^T A H Z a}{a^T S_* a} = \frac{\|A^{1/2} H Z S_*^{-1/2} b\|_2^2}{\|b\|_2^2}, \quad b = S_*^{1/2} a. \quad (\text{B.6})$$

Taking the supremum proves Eq. (B.5). The basis Z is used for analysis; the operator application does not construct these whitened coordinates.

Appendix B.1. Displacement magnitude and directional stiffness

The energy metric assigns different significance to displacement errors of equal magnitude. To make that dependence explicit, fix the retained rigid gauge by taking $q \perp R_P$, let $d = Fq - Eq$, and introduce a displacement norm induced by $W \succ 0$:

$$\delta_W(q) = \frac{\|d\|_W}{\|Eq\|_W}, \quad \mathcal{R}_{K,W}(v) = \frac{v^T K v}{v^T W v}, \quad \kappa_W(q, d) = \frac{\mathcal{R}_{K,W}(d)}{\mathcal{R}_{K,W}(Eq)}. \quad (\text{B.7})$$

For nonzero field error and nonzero reference energy, cancellation of $d^T W d$ and $u^T W u$, with $u = E q$, gives

$$\boxed{\varepsilon(q) = \delta_W(q)^2 \kappa_W(q, d).} \quad (\text{B.8})$$

Section 6.4 and Table ST04 use $W = \text{diag}(0, D)$ with $D = \text{diag}(K_{II})$, i.e. a Jacobi-weighted interior norm. This weighting is only positive semidefinite, but the identity still holds because d vanishes on the retained coordinates, and the rigid gauge is then not needed. The identity is a diagnostic decomposition of the energy error, not an independent explanation of it. The factor κ_W compares the stiffness content of the error and the equilibrium response; it is not a matrix condition number. An error concentrated in stiffer deformation than the loaded response has a large κ_W , making even a small relative displacement error mechanically significant. Thus an energy target ε_{tar} requires $\delta_W \leq \sqrt{\varepsilon_{\text{tar}}/\kappa_W}$. Correction must address this energy content as well as displacement magnitude.

Changing the displacement normalisation changes the corresponding amplification factor. An alternative based on the retained displacement is

$$\delta_P = \frac{\|d_I\|_2}{\|q\|_2}, \quad \kappa_P = \frac{d_I^T A d_I / \|d_I\|_2^2}{q^T S q / \|q\|_2^2}, \quad \varepsilon(q) = \delta_P^2 \kappa_P. \quad (\text{B.9})$$

Appendix C. Assembly and compliance ordering

For a global retained vector U , minimising the sum of substructure energies over their disjoint internal variables separates into the individual minimisations in Eq. (B.1). The minimised quadratic form is $\sum_m U^T B_m^T S_m B_m U$. This proves the equivalence of local condensation followed by assembly and elimination from the modular assembled system under the coupling assumptions in Section 2.3.

Let $\mathbb{D} = \hat{\mathbb{K}} - \mathbb{K}$. From Eq. (9),

$$U^T \mathbb{D} U = \sum_m \|H_m B_m U\|_{A_m}^2 \geq 0. \quad (\text{C.1})$$

If homogeneous supports make $\mathbb{K} \succ 0$, then both global systems are invertible and matrix inversion reverses their order. For any fixed nonzero load,

$$\hat{C} = f_g^T \hat{\mathbb{K}}^{-1} f_g \leq C = f_g^T \mathbb{K}^{-1} f_g. \quad (\text{C.2})$$

For completeness, the order reversal follows by setting $Q = \mathbb{K}^{-1/2} \hat{\mathbb{K}} \mathbb{K}^{-1/2} \succeq I$. Its eigenvalues are at least one, so $Q^{-1} \preceq I$; a congruence gives the inverse inequality. If every substructure also satisfies $0 \preceq \hat{S}_m - S_m \preceq \bar{\varepsilon} S_m$, applying the same argument to the upper and lower bounds gives

$$\mathbb{K} \preceq \hat{\mathbb{K}} \preceq (1 + \bar{\varepsilon})\mathbb{K}, \quad \frac{C}{1 + \bar{\varepsilon}} \leq \hat{C} \leq C, \quad \frac{C - \hat{C}}{C} \leq \frac{\bar{\varepsilon}}{1 + \bar{\varepsilon}}. \quad (\text{C.3})$$

The required local bound is uniform over directions. A mean over a finite bank has a different statistical meaning.

The assembled state error obeys the exact identity

$$\hat{U} - U = -\hat{\mathbb{K}}^{-1} \mathbb{D} U. \quad (\text{C.4})$$

Consequently, for a fixed finite assembly with uniformly stable support and bounded inverse, a family $H_m = O(t)$ gives $\mathbb{D} = O(t^2)$ and $\hat{U} - U = O(t^2)$ as $t \rightarrow 0$. This asymptotic statement holds with the exact discrete system fixed. Its constants depend on the assembly and do not prescribe which contribution dominates a finite-error sensitivity diagnostic.

Appendix D. Polynomial smoothing and coarse projections

For $0 < a < b$, the degree- k Chebyshev error polynomial and its energy bound are

$$p_k(t) = \frac{T_k((a + b - 2t)/(b - a))}{T_k((a + b)/(b - a))}. \quad (\text{D.1})$$

$$d_I^{\text{sm}} = p_k(D^{-1}A)d_I, \quad \|d_I^{\text{sm}}\|_A^2 \leq \rho_k^2 \|d_I\|_A^2, \quad \rho_k = \max_{\lambda \in \sigma(A)} |p_k(\lambda)|. \quad (\text{D.2})$$

Here $D = \text{diag}(A)$ and $\tilde{A} = D^{-1/2} A D^{-1/2}$.

Write $M = D^{-1}A$, $\theta = (a + b)/2$, $\eta_s = (b - a)/2$, and $\sigma = \theta/\eta_s > 1$. Starting from u_0 , keep its retained entries fixed and define the internal residual with the sign convention $r_j = (Ku_j)_I$. A recurrence producing the polynomial in Eq. (D.1) is

$$\begin{aligned}
\varrho_0 &= 1/\sigma, & s_0 &= -D^{-1}r_0/\theta, \\
u_{j+1} &= u_j + J_I^T s_j, & \varrho_{j+1} &= (2\sigma - \varrho_j)^{-1}, \\
s_{j+1} &= \varrho_{j+1}\varrho_j s_j - \frac{2\varrho_{j+1}}{\eta_s} D^{-1}r_{j+1}.
\end{aligned} \tag{D.3}$$

The last line is evaluated only when another step is needed. Zero steps return the initial field. To verify the polynomial, observe that $\varrho_j = T_j(\sigma)/T_{j+1}(\sigma)$, and apply $T_{j+2}(x) = 2xT_{j+1}(x) - T_j(x)$. The first error update is $d_1 = (I - M/\theta)d_0$, and the same recurrence gives $d_k = p_k(M)d_0$.

The matrices M and $\tilde{A} = D^{-1/2}AD^{-1/2}$ are similar, $p_k(M)$ is self-adjoint in the A -inner product, and

$$\|p_k(M)d\|_A^2 = \sum_j \lambda_j p_k(\lambda_j)^2 c_j^2, \quad c_j = v_j^T Dd. \tag{D.4}$$

This proves Eq. (D.2). If $a \leq \lambda \leq b$, the argument of T_k lies in $[-1, 1]$, where $|T_k| \leq 1$. For $0 < \lambda < a$, it lies between 1 and σ , where T_k increases from 1 to $T_k(\sigma)$. Hence $|p_k(\lambda)| \leq 1$ throughout $(0, b]$, so the polynomial is nonexpansive in the A -energy norm when the actual positive spectrum is contained in $(0, b]$. Modes below a can contract slowly, and eigenvalues above b can be amplified. The argument establishes contraction of the prescribed polynomial; it does not require every intermediate degree to improve monotonically.

The operational upper endpoint is estimated using a random power vector, 40 iterations of M , Euclidean normalisation, and a factor of 1.05. The correction implementation fixes the random generator seed and caches the interval for the geometry. The seed fixes the start vector for a given device type; the host evaluation in Table ST04 therefore uses a different start from the GPU runs. The separate smoothing diagnostic uses its own random start. The estimates are used to set the polynomial, while the result in Eq. (D.2) is conditional on its actual spectral values: a finite power estimate multiplied by a safety factor is an estimate of the endpoint, not the containment used above. For the 80 validation geometries we therefore compared the operational endpoint with $\lambda_{\max}(M)$ computed by Lanczos iteration with full reorthogonalisation on $D^{-1/2}AD^{-1/2}$ (78–150 steps; relative Ritz residual at most 9.6×10^{-4} , median 2.2×10^{-6}). The operational endpoint exceeds the converged value by 2.1–5.0% (median 3.9%) on every geometry, so the containment required by Eq. (D.2) holds for all evaluated cells. The strict

Gershgorin bound $\max_i \sum_j |A_{ij}|/A_{ii}$ is 4.7–19.4 times the operational endpoint (median 18.4); used as b , it would stretch the targeted interval by that factor and slow the contraction of the upper spectrum accordingly.

For a full-column-rank coarse basis V , with $A_c = V^T A V \succ 0$, define $Q_V = V A_c^{-1} V^T A$. Direct multiplication gives $Q_V^2 = Q_V$ and $Q_V^T A = A Q_V$. Thus Q_V is an A -orthogonal projection and $C_V = I - Q_V$ is its complementary projection. With $b_r = V^T A d$,

$$\|d\|_A^2 = \|C_V d\|_A^2 + \|Q_V d\|_A^2, \quad \|Q_V d\|_A^2 = b_r^T A_c^{-1} b_r. \quad (\text{D.5})$$

This proves Eq. (16). Applying Eq. (D.2) before and after this projection yields $\|P_k C_V P_k d\|_A \leq \rho_k^2 \|d\|_A$. Because $\rho_k < 1$ whenever the positive spectrum lies in $(0, b]$, the cycle is an energy contraction and yields the operator ordering in Eq. (17). For the spectra of the detailed cells ($\lambda_1 \approx 4\text{--}6 \times 10^{-4}$ against $a \approx 0.17$), however, $\rho_8^4 \geq 0.98$, so the estimate guarantees little more than non-expansion; it uses no approximation property of the coarse space, and the reductions by one to two orders of magnitude in Section 6.5 are empirical. Repeating the cycle cannot increase the energy error; changing the smoothing degree changes the polynomial and is not guaranteed to be monotone. A two-grid convergence estimate would require an approximation property of the Q_1 space for walls about one element thick Xu & Zikatanov (2002). This is the standard energy-projection mechanism of subspace correction Xu (1992).

Appendix E. Transpose of the complete extension

All transposes below use the Euclidean pairing of the stored displacement and nodal-force coordinates. Write $M_I = J_I^T J_I$. Transposing Eq. (5) gives

$$\hat{E}^T y = J_P y + C_R^T R^T M_I y + \Pi_P^T \mathcal{N}_\theta(\eta)^T M_I y. \quad (\text{E.1})$$

The overwrite therefore selects the direct retained contribution and masks the field sent through the network transpose. The geometry coefficients are held fixed during this displacement transpose.

Set $s_k(t) = [1 - p_k(t)]/t$, with the polynomial continuation at zero. The full-field smoothing map $\mathcal{T}_k$ has the block representation

$$\mathcal{T}_k = \begin{bmatrix} I_p & 0 \\ -s_k(M)D^{-1}K_{IP} & p_k(M) \end{bmatrix}, \quad \mathcal{T}_k^T y = \begin{bmatrix} y_P - K_{PI}s_k(M)D^{-1}y_I \\ Dp_k(M)D^{-1}y_I \end{bmatrix}. \quad (\text{E.2})$$

Indeed, the final internal field is its equilibrium value plus $p_k(M)$ times the initial error. Replacing $(I - p_k(M))A^{-1}$ by $s_k(M)D^{-1}$ yields the first expression. The second follows from $M^T D = DM$, which implies $p_k(M)^T = Dp_k(M)D^{-1}$ and the corresponding identity for s_k . Although the forward map preserves retained displacement, its transpose contributes a retained force through the upper block in Eq. (E.2).

For the full-field coarse correction,

$$\mathcal{W}_c = I_{n_a} - J_I^T V A_c^{-1} V^T J_I K, \quad \mathcal{W}_c^T = I_{n_a} - K J_I^T V A_c^{-1} V^T J_I. \quad (\text{E.3})$$

A cycle $\mathcal{W} = \mathcal{T}_{\text{post}} \mathcal{W}_c \mathcal{T}_{\text{pre}}$ therefore uses the reverse sequence $\mathcal{W}^T = \mathcal{T}_{\text{pre}}^T \mathcal{W}_c^T \mathcal{T}_{\text{post}}^T$. The complete reaction is

$$\hat{S}q = \hat{E}^T \mathcal{W}^T K \mathcal{W} \hat{E}q. \quad (\text{E.4})$$

The same formulas hold for the symmetric shifted inverse defined below when it replaces the coarse inverse consistently in both actions. Iteration counts, coarse bases, and polynomial coefficients depend on geometry and remain fixed across input directions. An input-dependent iterative stopping rule would require a separate analysis of linearity and differentiation.

Appendix F. Coarse spaces, factorisation, and arithmetic

Appendix F.1. Explicit coarse-space study

Coarse tensor-product shape functions are evaluated at the active background-node coordinates and restricted to I . Standard Q_1 and Q_2 vector spaces carry three displacement coefficients per coarse node. The enriched partition-of-unity space carries 12 coefficients per vertex through fields of the form $\sum_v N_v(x)[a_v + B_v(x - x_v)]$, with $a_v \in \mathbb{R}^3$ and $B_v \in \mathbb{R}^{3 \times 3}$. The spaces examined use Q_1 grids of 9, 17, or 33 vertices per axis, a Q_2 grid of 17 nodes per axis, and enriched Q_1 grids of 9 or 17 vertices per axis.

Columns with an absolute interior-support sum at most 10^{-14} are removed. In the fixed-weight CPU study, the symmetric internal matrix is

formed from its stored upper triangle. The coarse matrix is explicitly computed as $V^T AV$ and symmetrised. Columns whose diagonal energy is at most 10^{-12} times the largest coarse diagonal are then removed. The resulting sparse matrix is factorised with PARDISO, or SuperLU when PARDISO is unavailable. This construction adds no explicit diagonal shift. The projection formulas in Eqs. (16)–(17) apply to linearly independent surviving coarse columns and the exact Galerkin action. Interpreting a recorded numerical solve through those formulas additionally requires verification of its solve accuracy.

The enriched generating functions are linearly dependent, and the surviving columns are not certified independent; the enriched entries of Table ST07 are therefore reported as numerical observations in Supplementary Note S3, and the $Q_1(17)$ result is the principal coarse-correction result.

For each input direction, the study evaluates the exact field and the uncorrected network field once. It compares the network alone, a smoothing tail, a coarse update, coarse followed by smoothing, smoothing on both sides of the coarse update, and a zero-interior initialization followed by the same complete cycle. Its energy ratios use a recomputed teacher energy for the supplied direction. The fields are converted to float64 before correction.

Appendix F.2. Differentiable correction implementation

The differentiable implementation constructs coarse support metadata and recovers a coarse matrix using coloured stiffness probes. Its configured stencil reach is four fine-grid node spacings. With h_g fine-node spacings per coarse-vertex spacing, the probe radius is $R_g = 2 + \lceil 4/h_g \rceil$; colours use vertex coordinates modulo $2R_g + 1$ and the component or enrichment slot. Recovery assumes the resulting colour separation resolves every interacting coarse pair. This setup supports the Q_1 and enriched Q_1 spaces. The explicit Q_2 study uses the CPU construction in Appendix F.1.

The recovered matrix is Jacobi scaled with the corresponding column scaling of V . Cholesky factorisation reads its lower triangle, trying diagonal shifts in the order $0, 10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}$. The selected value is stored with the geometry factorisation; in the recorded setups it was 0 on six cells and 10^{-12} on H1 (`evidence/time_setup.json`). On three lattice cells, the probed factor agrees with the element-assembled Galerkin factor to a relative 1.3×10^{-11} or better (record `evidence/coarse_test.jsonl`). To characterize such a solve algebraically, let V_s denote the scaled basis,

$H_c = V_s^T A V_s \succeq 0$, $\xi \geq 0$, $H_c + \xi I \succ 0$, $B_c = (H_c + \xi I)^{-1}$, and $b_s = V_s^T r_I$. Then

$$\|d_I\|_A^2 - \|d_I - V_s B_c b_s\|_A^2 = b_s^T B_c (H_c + 2\xi I) B_c b_s \geq 0. \quad (\text{F.1})$$

Expanding the squared norms, as in Eq. (J.9) with basis V_s , $G = B_c$, and A_c replaced by H_c , gives $b_s^T (2B_c - B_c H_c B_c) b_s$, and multiplication by $H_c + \xi I$ establishes the equality. A positive shift changes the exact projection but preserves this energy decrease when the solved matrix is the stated scaled Galerkin matrix. The identity assumes that matrix equality, symmetry, and consistent forward and transpose solves.

Appendix F.3. Precision and normalisation

The learned displacement extension, rigid-body reconstruction, and prescribed retained entries use float32. Quadratic energies and sparse stiffness products use float64. The differentiable correction implementation computes its relaxation and coarse algebra in float64 in training and in the accuracy studies of Sections 6.2–6.9, and returns to the input displacement dtype; the timed lattice route of Table 5 runs the smoothing and the coarse solve in float32 (Supplementary Note S5). Convolutions use true float32 arithmetic; TF32 and lower precisions are not used. The explicit inference transpose also crosses the float64/float32 boundary around its network action. Sensitivity contractions use float32 element products with float64 accumulation. These arithmetic choices approximate the real linear maps in the preceding derivations.

Evaluation path	Initial field and correction	Condensed action
Fixed-weight coarse study (base network)	Float32 network field converted to float64; explicit CPU Galerkin matrix and float64 correction	The saved comparison evaluates corrected field energies; it is separate from the deployment timing study.

Evaluation path	Initial field and correction	Condensed action
Differentiable correction (training; accuracy evaluation of NICE, NICE-post and Smoothing-trained)	Float32 learned field; relaxation and coarse algebra in float64; correction routines return to their input dtype	The correction and its complete transpose enter the stated variational action. Smoothing-trained uses eight smoothing steps and no coarse solve.
Timed lattice route (Table 5)	Float32 network and rigid reconstruction; smoothing and coarse solve in float32; float64 stiffness product of the condensed action	Same operations and transpose as above; the fp32 correction changes the $2 \times 2 \times 2$ compliance errors by less than 3×10^{-9} (Supplementary Note S5).
Explicit inference implementation	Float32 network and rigid reconstruction; float64 stiffness product; dtype conversions before the transpose action	The complete transpose follows the actual configured correction sequence, with finite-precision consistency assessed separately.

In the mixed-precision two-grid implementation, each smoothing routine returns its input dtype. Consequently a float32 pre-smoothed field can be rounded before entering the float64 coarse solve; the final corrected field returns to the original dtype. These casts belong to the numerical implementation rather than the real-linear maps used in the identities.

The role of actual retained values can be isolated algebraically. Suppose a computed field v has retained entries $q' = J_P v = q + b_P$, and set $d = v - Eq$. Without assuming $b_P = 0$,

$$v^T K v - q'^T S q' = d^T K d + 2b_P^T S q. \quad (\text{F.2})$$

Using the actual trace instead gives the nonnegative variational difference $v^T K v - q'^T S q'$. In exact arithmetic applied to those actual vectors, with $r_I = (Kv)_I$,

$$v^T K v - 1 = r_I^T A^{-1} r_I + (q'^T S q' - 1). \quad (\text{F.3})$$

Thus a score formed by subtracting one also depends on the normalisation of the stored direction. Recomputed teacher denominators, stored unit-energy assumptions, and finite-precision quadratic forms are separate parts of a numerical energy diagnostic.

Appendix F.4. Moment integration

The teacher's standard entry points initialize 4^3 subcells per active background element and refine partially occupied subcells once into 2^3 children. Thus the finest subcell width near the material boundary is $h/8$. Fully occupied subcells contribute analytic tensor-product monomial moments. At the finest partial level, each subcell is divided into six Kuhn tetrahedra and clipped successively by the three interpolated inequalities defining the band and the macro cut. The tetrahedral quadrature helper is called with its order parameter set to four. The accumulated moments are multiplied by the physical Jacobian $(2n)^{-3}$. Refinement is local to partial subcells; it does not uniformly subdivide every active element to the finest level.

Appendix G. Network coefficients and directional training

Appendix G.1. Geometry features and displacement maps

The architecture in Figure 3 separates geometry-dependent coefficient generation from displacement propagation. Let N denote the number of active nodes, N_P the number of retained nodes and B the number of simultaneous displacement directions. Then $n_a = 3N$, $p = 3N_P$, and the deformational input $q_d = \Pi_P q$ has shape $p \times B$. The fine displacement features have shape $N \times B \times 32$; a coarse latent level with N_ℓ participating vertices has shape $N_\ell \times B \times 32$. Geometry embeddings have 64 components and carry no displacement-direction dimension. Element, face, retained-node and grid-transfer incidences are constructed before the network is evaluated.

Geometry encoding. Each element has 126 input features: material volume fraction, its logarithm, and the remaining 124 moments normalised by material volume. The 11 node features contain retained, box, cut-band and weak-support indicators, a normalised logarithmic stiffness diagonal, and six sine/cosine coordinate entries. The stiffness summary is the Frobenius norm of the node's diagonal 3×3 block. A node is designated weakly supported when this norm is less than 0.01 times its median over active nodes.

These features supply information about both material occupancy and its mechanical support.

The element encoder maps 126 inputs to a 64-component embedding. Mean aggregation of incident element embeddings, concatenated with the 11 node features, supplies the 75-input node encoder. Two residual message-passing rounds then update elements from the mean embeddings of their 27 nodes and update nodes from their incident elements. Each update uses the current embedding and the aggregated neighbouring embedding as a 128-component input. A ghost-face embedding is formed from the two adjacent element embeddings and a three-component indicator of the face axis. All encoders and coefficient heads use two affine layers separated by GELU, with hidden width 64. Every reported predictor uses the base features described here; the optional five-feature extension is disabled.

Local linear interactions. The three retained displacement components are lifted by a shared matrix $W_{\text{in}} \in \mathbb{R}^{3 \times 32}$, while internal features are initially zero. Denote this initial feature field by $X^0(q)$, with its dependence on q occurring through q_d . For each element or face stencil t , local node slot s and head h , the interaction first gathers and mixes channels:

$$Z_{th} = \left(\sum_{s=1}^{27} a_{tsh}(\eta) X_{i(t,s)} \right) W_{\ell h}, \quad \Delta X_i = \sum_{(t,s): i(t,s)=i} \sum_{h=1}^4 \omega_{ts} b_{tsh}(\eta) Z_{th}.$$

Here $W_{\ell h} \in \mathbb{R}^{32 \times 32}$ is a shared learned channel map for layer ℓ and head h . The scalar gather and scatter coefficients a and b are generated separately from the element or face embedding, the incident node embedding and an eight-dimensional slot embedding, a learned 27×8 table indexed by the local slot (one table for element stencils and one for face stencils). Their 136-component concatenation identifies both the local material context and the position within the stencil. The four heads provide distinct gather--mix--scatter contributions on the same fixed incidence. Equation (4) adds their update to the incoming state and restores the prescribed retained features after every local interaction.

The element stencil contains all 27 Q_2 nodes. The learned ghost-face stencil contains 18 owner-side nodes and nine neighbour-side nodes, ordered consistently by face direction. It defines a learned communication map on a face neighbourhood; the full mechanical ghost-penalty contribution remains in K .

The node scatter accounts for the number and relative stiffness of incident contributions. If k_{es} is the Frobenius norm of the element's diagonal 3×3 block at slot s , the element weights are

$$\pi_{es} = \frac{k_{es}}{\sum_{(e',s') \mapsto i} k_{e's'}}, \quad \omega_{es} = \frac{1 - \lambda_{\text{mix}}}{\deg(i)} + \lambda_{\text{mix}} \pi_{es}, \quad (e, s) \mapsto i. \quad (\text{G.1})$$

The face weights use the corresponding ghost-contribution diagonal blocks and face incidences. When all incident stiffness norms at a node vanish, the stiffness share is defined by inverse incidence count. Each layer has its own learned λ_{mix} , initialised at zero (plain incidence averaging), without a convex-interval constraint. This weighting changes coefficients on the prescribed scatter map. For the additional weak-region layers, a stencil is selected when it contains a weak node; its weights retain the normalisation of the full element or face incidence set.

Multilevel propagation. The latent hierarchy enlarges the spatial range of communication while retaining the fine field through additive skips. Let t_{ia} be the fixed trilinear incidence weight between a fine node i and a coarse vertex a . Positive geometry coefficients r_i and p_i define restriction and prolongation componentwise:

$$(\mathcal{R}X)_a = \frac{\sum_i t_{ia} r_i X_i}{\sum_i t_{ia} r_i}, \quad (\mathcal{P}X_c)_i = p_i \frac{\sum_a t_{ia} X_{c,a}}{\sum_a t_{ia}}.$$

Only coarse vertices reached by the stored trilinear incidences participate. The denominators depend on geometry alone, so both maps are linear in displacement. Restriction and prolongation are separately parameterised. Their relationship does not impose symmetry on the raw extension; the transpose in Eq. (6) is obtained from the complete composition of operations.

On each coarse level, a residual convolution takes the form

$$X \leftarrow X + g_{\ell_j}(\eta) \odot \text{Conv}_{\ell_j}(X).$$

Each convolution has a $3 \times 3 \times 3$ kernel, 32 input and output channels, unit padding and no bias or displacement activation. Participating latent nodes are inserted into the corresponding background grid, with unused positions set to zero; the convolution output is sampled back at those nodes. The geometry branch supplies one multiplicative coefficient per participating node, channel and convolution. Its coarse embeddings are obtained by fixed trilinear weighted averaging.

The full displacement sequence is specified below. Grid sizes count background positions per axis, whereas the participating node counts depend on geometry. Every row carries 32 displacement channels until the final projection.

Stage	Operations in execution order
Fine input, 65 grid	Lift q_d on retained nodes, set internal features to zero; apply four element-then-face interaction pairs; save S_0 .
Downward, 33 grid	Restrict 65→33; apply two residual convolutions; save S_1 .
Downward, 17 grid	Restrict 33→17; apply two residual convolutions; save S_2 .
Downward, 9 grid	Restrict 17→9; apply two residual convolutions.
Upward, 9 grid	Apply two residual convolutions; prolong 9→17; add S_2 .
Upward, 17 grid	Apply two residual convolutions; prolong 17→33; add S_1 .
Upward, 33 grid	Apply two residual convolutions; prolong 33→65; add S_0 .
Fine output, 65 grid	Restore retained features; apply four element-then-face interaction pairs, then four weak-region element-then-face pairs; project 32→3.

At each upward step the combination is $X = S_\ell + \sigma_\ell \mathcal{P}_\ell X_c$, with a learned scalar σ_ℓ . Thus convolution precedes prolongation, the skips are additive, and the 9-position grid receives two downward and two upward convolutions. There are twelve coarse convolution updates in total. The fine retained values are restored after the hierarchy returns to the active nodes. These latent operations communicate within the extension; the mechanical operator still accepts and returns all p retained coordinates. The Galerkin correction of Section 5 instead uses the internal stiffness and equilibrium residual.

Coefficient bounds and output. The local heads produce separate gather/scatter values for eight ordinary element layers, twelve face layers and four additional weak-region element layers, each with four heads and 27

slots. Raw head outputs are multiplied by 0.2. For a finite positive group bound $a_{\max}$, the coefficient map is

$$\mathcal{C}(z) = \begin{cases} z, & |z| \leq k_b a_{\max}, \\ \text{sgn}(z) \left[k_b a_{\max} + (1 - k_b) a_{\max} \tanh\left(\frac{|z| - k_b a_{\max}}{(1 - k_b) a_{\max}}\right) \right], & |z| > k_b a_{\max}. \end{cases}$$

All predictors of Table 2 set $k_b = 0.5$; each $a_{\max}$ is twice the largest magnitude of its coefficient group recorded in a calibration pass over training geometries, so the map is the identity up to that recorded maximum. Table ST20 (Supplementary Note S7) lists the groups, the initialisation and the parameter count of every block. An infinite bound denotes the identity map. Bounds are fixed model-state arrays indexed by coefficient type, layer and gather/scatter role. Positive transfer coefficients apply the corresponding bound to $\text{softplus}(z)$ and add 10^{-3} ; convolution coefficients use $2 \text{sigmoid}(z)$. All these nonlinearities act on geometry-derived coefficients. The channel matrices, convolution kernels and skip scalars are shared learned parameters, whereas the coefficient-head outputs vary with geometry.

A bias-free matrix $W_{\text{out}} \in \mathbb{R}^{32 \times 3}$ returns the fine features to nodal displacement in the prescribed coordinate order. Rigid reconstruction and retained-value restoration then produce Eq. (5). The energy action Eq. (6) uses this complete map, together with the prescribed correction, and its transpose; it therefore determines the force from the same displacement representation.

Frame consistency. For an orthogonal cube transformation, the deterministic signed permutation maps satisfy

$$K' = T_a K T_a^T, \quad E' = T_a E T_P^T, \quad S' = T_P S T_P^T. \quad (\text{G.2})$$

The augmented geometry and vector field are passed through the network and the predicted displacement is returned to the original frame for mechanical evaluation. These relations describe the transformed target and the frame-consistency condition. Sampling the cube views trains the geometry-conditioned extension towards that condition; displacement linearity alone does not establish rotational equivariance.

Appendix G.2. Direction banks

The normalised retained directions are

$$q_j = \frac{\Pi_P \tilde{q}_j}{\sqrt{(\Pi_P \tilde{q}_j)^T S (\Pi_P \tilde{q}_j)}}. \quad (\text{G.3})$$

The direction classes used by the trainer have the following mechanical definitions.

Class	Construction before rigid removal and energy normalisation
<code>force</code>	Equilibrated nodal loads from smooth plane waves or localized Gaussian patches, with a subset loading the cut boundary.
<code>force_c</code>	Consistently integrated self-equilibrated tractions on material box faces; a subset also loads the cut face. Equilibrium is imposed in traction-quadrature space.
<code>face / face_c</code>	Single-face equilibrated nodal loads or consistently integrated tractions.
<code>support</code>	Responses with soft spring support on one box face and equilibrated loads on other faces.
<code>support_k</code>	Consistent loads with springs scaled by the local stiffness diagonal and a logarithmically sampled factor in $[0.3, 3]$; the cut face remains unloaded.
<code>macro</code>	Equal sampling of uniform strain, quadratic, and cubic imposed displacement fields.
<code>grf</code>	Multiscale displacement fields spanning 0.5–24 spatial cycles across the reference box.
<code>glued</code>	Traces induced by a neighbouring cell with shared interface degrees of freedom and far-face clamping or springs; one quarter of samples load only the neighbour.

Class	Construction before rigid removal and energy normalisation
adv	Directions selected by the generalised-energy search described below.

The isolated teacher removes the rigid component and normalizes the resulting retained direction by Eq. (G.3). Glued responses therefore supply a contextual direction while the energy and sensitivity labels refer to the target cell's operator. Banks store the normalised vectors in float32.

Appendix G.3. Optimisation and difficult-direction search

The predictors of Table 2 use batches of 16 directions, update the geometry every step, keep three geometries in the device pool, and replace a pool entry every 100 steps. Their nominal class weights, in the order **force**, **force_c**, **support**, **support_k**, **face**, **face_c**, **macro**, **grf**, **glued**, **adv**, are 0.10, 0.10, 0.075, 0.075, 0.10, 0.10, 0.10, 0.125, 0.075, 0.15. Weights are renormalized over available classes and assigned by systematic quotas. A run of N updates therefore visits at most $N/100 + 3$ distinct geometries of its training pool: 305 geometries for the base network and 591 for the continued predictors, as recorded at the start of each run (Table ST01).

Optimisation uses Adam, a one-cycle learning-rate schedule with peak 3×10^{-4} , 5% warm-up, cosine decay, and final division factor 100. The gradient norm is clipped at one. Selected weights use an exponential moving average with decay 0.9997 and zero-initialization bias correction. Table 2 identifies the model roles, and Table ST01 gives the training and weight-selection settings for each arm. The energy term in Eq. (8) uses a 10^{-12} lower clamp inside the logarithm; the reported sensitivity loss is the squared relative eight-corner norm with unit weight. The optional bank-spectrum loss has zero default weight in these configurations.

NICE and the smoothing-trained predictor continue the base network from its selected weights for 15,000 steps with the same schedule, class weights, training pool and seed as the uncorrected continuation, so that the three continuations differ only in the correction used in the forward map of Eq. (8): none for the uncorrected predictor, eight Chebyshev steps for the smoothing-trained one, and the eight-step, $Q_1(17)$, eight-step sequence for NICE. The correction is recomputed for every geometry of the device pool

(smoothing interval by power iteration, coarse matrix by coloured probing, Appendix F.2) and is treated as a fixed linear map of the network output. Gradients of Eq. (8) are propagated through the smoothing recurrence and the coarse solve by automatic differentiation in double precision; the transposes used at evaluation (Appendix E) are the explicit adjoints of the same operations. Checkpoints were scored at 7,500 and 15,000 updates on the validation list of Section 6.1 (views 0 and 17); the 15,000-update weights are reported for every continuation, and for the uncorrected predictor and NICE they scored lower. The base network's weights were selected among four checkpoints (10,000–40,000 updates; Table ST01).

The adversarial search initializes eight candidate directions and performs four block iterations when a geometry is loaded. Applying $S_{\perp}^{-1}$ uses an equilibrated Neumann solve: six deterministically selected displacement pins remove rigid freedom, and the result is projected into the retained rigid complement. A bank search instead forms $G = Q_b^T S Q_b$ and $\hat{G} = (F Q_b)^T K (F Q_b)$, solves a generalised eigenproblem after flooring the search metric at $10^{-3} \lambda_{\max}(G)$, and renormalizes selected candidates with the original G . The floor belongs to the search coordinates. Candidate energy ratios remain directional observations; their maximization does not supply an all-direction upper bound.

Appendix H. Sensitivity identities and design intervals

The exact condensed derivative and the fixed-displacement energy derivative are

$$S_{,c} = E^T K_{,c} E, \quad \frac{\partial}{\partial \tau_c} \left(\frac{1}{2} q^T S q \right) = \frac{1}{2} (E q)^T K_{,c} (E q). \quad (\text{H.1})$$

For an assembly, the full local field difference is

$$F \hat{q} - E q = (F - E) q + E (\hat{q} - q) + (F - E) (\hat{q} - q). \quad (\text{H.2})$$

On a differentiable interval with fixed active topology and coordinate maps, differentiating $J_P E = I_p$ gives $J_P E_{,c} = 0$. Since $J_I K E = 0$, both terms $E_{,c}^T K E$ and $E^T K E_{,c}$ vanish. Differentiating $S = E^T K E$ therefore proves Eq. (H.1).

For a supported equilibrium, differentiating $\mathbb{K} U = f_g$ at a fixed load gives $U_{,c} = -\mathbb{K}^{-1} \mathbb{K}_{,c} U$ and hence $C_{,c} = -U^T \mathbb{K}_{,c} U$. With fixed gathers and

Eq. (H.1), this becomes the sum of $-u_m^T K_{m,c} u_m$ over affected cells. For an isolated energy-normalised direction, this label holds the base-design force fixed. The assembled sensitivity labels likewise hold the base-design nodal load fixed, including loads generated from consistent tractions. Reassembling a prescribed surface traction over a changing material boundary introduces the load-derivative term stated below; differentiating the direction normalisation also defines a different quantity.

Expanding the field-based quadratic estimate at $u + d$ proves Eq. (13). In the Euclidean norm, it implies

$$|\tilde{s}_c - s_c| \leq 2\|K_{,c}u\|_2\|d\|_2 + \|K_{,c}\|_2\|d\|_2^2. \quad (\text{H.3})$$

For the corner-thickness band in Eq. (1), the nonnegative trilinear shape functions make the material domains nested as one thickness parameter increases. With fixed basis functions and ghost stabilisation, exact integration gives $K_{,c} \succeq 0$ and nonpositive equilibrium compliance derivatives; Eq. (J.6) of Appendix J.5 proves the semidefiniteness; a numerical derivative inherits it only while its quadrature preserves the nesting of the material domains, which the discrete moments of this study do not do in a few elements (Appendix J.5). The linear cross term in Eq. (13) can have either sign, while the quadratic term is nonpositive; their relative importance depends on derivative-weighted alignment and error amplitude. General design parameterizations can instead produce indefinite derivative matrices. The numerical moment-difference implementation is specified below. In an assembly, substituting Eq. (H.2) into the same quadratic expansion includes both the change of trace and its interaction with the extension error. Field-only and solution-only relative errors are therefore not additive scalar contributions.

For the surrogate operator, direct differentiation gives

$$\hat{S}_{,c} = F_{,c}^T K F + F^T K_{,c} F + F^T K F_{,c}. \quad (\text{H.4})$$

Apply the equilibrium compliance derivative to the assembled surrogate and use $J_P F_{,c} = 0$. The two extension terms become $2(F_{I,c}\hat{q})^T r_I$, proving Eq. (14). Its magnitude is bounded by

$$|\hat{C}_{,c} - \tilde{s}_c| \leq 2\|F_{I,c}\hat{q}\|_A\|r_I\|_{A^{-1}} \quad (\text{H.5})$$

for a single affected cell, with a sum of such bounds for several cells. This bound involves the extension's design derivative as well as its equilibrium

residual. It provides no fixed error order in the field amplitude without a corresponding assumption on that derivative. If the load depends on design, the full compliance derivative also contains $2f_{g,c}^T \hat{U}$, and correspondingly $2f_{g,c}^T U$ for the exact assembly; changes of coordinate, assembly, or support maps contribute their own chain-rule terms, including the derivatives of $B_m^T S_m B_m$ when B_m varies. Fixed active topology does not, on its own, imply differentiability of quadrature branches or a moving-load map.

In the teacher, moment derivatives are approximated by

$$M_{e\alpha,c} \approx \frac{M_{e\alpha}(\boldsymbol{\tau} + h_c e_c) - M_{e\alpha}(\boldsymbol{\tau} - h_c e_c)}{2h_c}, \quad h_c = 10^{-5} \tau_c. \quad (\text{H.6})$$

The active set is fixed for this calculation, and the elasticity templates and ghost contribution are held constant. The moment quadrature is recomputed at each perturbed thickness, so its clipping and integration branches may change. The resulting matrices represent a numerical design derivative on that prescribed topology. Training differentiates the field-based sensitivity loss with respect to the predicted field and network parameters. Computing the complete design derivative in Eq. (14) additionally differentiates the geometry-dependent extension and correction maps.

Appendix I. Residual lower diagnostics

For any nonzero internal test vector z , Cauchy–Schwarz applied to $A^{1/2}z$ and $A^{-1/2}r_I$ gives

$$\Delta(q) := q^T(\hat{S} - S)q = r_I^T A^{-1} r_I \geq L_z := \frac{(z^T r_I)^2}{z^T A z}. \quad (\text{I.1})$$

A preconditioned residual, a coarse approximation, or a Krylov iterate can supply z . Evaluating the final one-vector quotient with the original A makes the inequality independent of how that vector was generated. For $e_h = \hat{u}^T K \hat{u}$ and $e_h - L_z > 0$, the map $x \mapsto x/(e_h - x)$ is increasing on the relevant interval. Therefore

$$\varepsilon(q) \geq \frac{L_z}{e_h - L_z}, \quad \mu_* \geq 1 + \frac{L_z}{e_h - L_z}. \quad (\text{I.2})$$

If the exact reference energy is known, $L_z/(q^T S q)$ gives a sharper lower estimate. A large value detects an inaccurate direction; a small value alone

does not bound the error from above. These are real-arithmetic inequalities, with numerical evaluation requiring the stated original quadratic forms.

The reduced-boundary ablation of Section 6.8 restricts the retained box-face displacements instead of the interior; its restricted Galerkin system and conditions are given in Supplementary Note S4.

Appendix J. Further variational and mechanical analysis

This appendix uses the notation of the main text and preceding appendices to develop the variational analysis of the prescribed discrete system. Analytical matrix examples in Appendix J.9 illustrate the resulting error orders and mechanical relationships.

Appendix J.1. Assumptions and mechanical target

In this appendix, $f \equiv f_g$ denotes the assembled retained load.

For each cell, $K = K^T \succeq 0$, the retained and internal coordinate sets P, I are fixed, and $A = K_{II} \succ 0$. The exact extension is defined by $E_P = I$ and $E_I = -A^{-1}K_{IP}$. A linear extension F has $F_P = I$. Set $H = F_I - E_I$, so $F = E + J_I^T H$, $S = E^T K E$, and $\hat{S} = F^T K F$. All transposes include the entire extension and correction. Interior body loads are zero; equivalently, the load functional factors through the retained coordinates. Consistent boundary loads have this property when all coefficients with nonzero boundary virtual work are retained.

The cell interiors are disjoint, all intercell couplings act through the retained coordinates, and the assembly maps B_m include fixed homogeneous supports. The supported exact assembled stiffness $\mathbb{K} = \sum_m B_m^T S_m B_m$ is positive definite. The same local stabilised matrices define both the reference and surrogate assemblies. The applied global load $f \neq 0$ is fixed. These assumptions distinguish the modular reference from a different monolithic discretisation that introduces additional intercell stabilisation.

Design statements additionally use a differentiable interval on which the active coordinates, retained coordinates, assembly maps, supports, and load are fixed. Stiffness derivatives include every varying term in the chosen discrete specification. In the present frozen sensitivity implementation, moment differences vary the body stiffness while the selected ghost matrix and its coefficient are fixed. The algebraic field-error expansions also hold for the saved symmetric finite-difference derivative matrices; interpreting them as

exact design derivatives requires convergence to the derivative of that discrete system.

The full retained nodal representation need not be a minimal independent basis of surface traces. The cut band keeps the active coefficients needed for the cut-surface load functional. This sufficiency, and containment of all intercell couplings in P , is the property used by condensation and assembly.

Appendix J.2. Why the complete transpose gives a quadratic operator error

Appendix B derives $\hat{S} - S = H^T A H$ and $r_I = (K F q)_I = A H q$ from the expansion in Eq. (B.1), in which internal stationarity $J_I K E = 0$ removes both cross terms. The stabilised discrete-energy excess at fixed q is one half of $q^T H^T A H q$; the reported energy excess uses the quadratic energy without the factor one half.

The retained block of the unbalanced discrete force, used alone, would be

$$(K F)_P = S + K_{PI} H.$$

The additional variational reaction is $F_I^T (K F)_I = (E_I + H)^T A H$. Since $E_I^T A = -K_{PI}$, its term $E_I^T A H$ cancels $K_{PI} H$. Thus

$$(K F)_P + F_I^T (K F)_I = S + H^T A H. \quad (\text{J.1})$$

The complete transpose supplies both reciprocity and cancellation of the first-order stiffness error. A retained-force extraction without this term generally has an $O(H)$ error and need not be symmetric.

Under $\ker K = \text{range } R$, full column rank of R_P , and rigid reproduction $F R_P = R$, Eqs. (B.2) and (B.4) give $\ker \hat{S} = \text{range } R_P$ with two-sided annihilation. The conclusion excludes extra modes without adding stiffness. Positive semidefiniteness alone does not imply rigid reproduction; rigid reproduction alone does not exclude extra modes in a reference K with a larger kernel.

For any global vector, the quadratic error is the sum of nonnegative local errors in Eq. (C.1). Consequently $\hat{\mathbb{K}} \succeq \mathbb{K} \succ 0$. This also proves supported solvability even if some local cells are only semidefinite. A high condition number can still increase the cost and sensitivity of the numerical solve.

Appendix J.3. Compliance as the full reconstructed error energy

Let $U = \mathbb{K}^{-1} f$, $\hat{U} = \hat{\mathbb{K}}^{-1} f$, $q_m = B_m U$, $\hat{q}_m = B_m \hat{U}$, $u_m = E_m q_m$, and $\hat{u}_m = F_m \hat{q}_m$. Write $\Delta = \hat{\mathbb{K}} - \mathbb{K} = \sum_m B_m^T H_m^T A_m H_m B_m$, the matrix $\mathbb{D}$ of Eq. (C.1).

For conforming cell fields, use the energy form $a(v, v) = \sum_m v_m^T K_m v_m$. Shared retained coefficients are counted through their separate cell energy contributions. Exact local equilibrium and the global equation give $a(u, v) = f^T V_P$, where V_P denotes the free assembled trace. At the surrogate equilibrium, $a(\hat{u}, \hat{u}) = f^T \hat{U} = \hat{C}$. Hence

$$C - \hat{C} = a(\hat{u} - u, \hat{u} - u) = \|\hat{U} - U\|_{\mathbb{K}}^2 + \sum_m \|H_m \hat{q}_m\|_{A_m}^2.$$

The second equality follows from $\hat{u}_m - u_m = E_m(\hat{q}_m - q_m) + J_{I,m}^T H_m \hat{q}_m$. The two terms are K_m -orthogonal because $J_{I,m} K_m E_m = 0$. Thus compliance underestimation measures the total energy error of the reconstructed field, whereas the retained-solution error accounts for only one part of it. In particular,

$$\frac{\|\hat{U} - U\|_{\mathbb{K}}}{\|U\|_{\mathbb{K}}} \leq \sqrt{\frac{C - \hat{C}}{C}}.$$

This nonasymptotic bound is valid without a small-error assumption. The sharper asymptotic retained-solution order is derived below.

Define $T = \mathbb{K}^{-1/2} \Delta \mathbb{K}^{-1/2} \succeq 0$, $z = \mathbb{K}^{-1/2} f$, $\rho = \|T\|_2$, and

$$\beta = \frac{U^T \Delta U}{C} = \sum_{m: q_m^T S_m q_m > 0} w_m \varepsilon_m(q_m), \quad w_m = \frac{q_m^T S_m q_m}{C}.$$

For zero-energy cells with exact rigid reproduction, the corresponding numerator is also zero and its contribution is defined directly as zero. Spectral calculus gives

$$\frac{C - \hat{C}}{C} = \frac{z^T T (I + T)^{-1} z}{z^T z}, \quad \frac{\beta}{1 + \rho} \leq \frac{C - \hat{C}}{C} \leq \frac{\beta}{1 + \beta} \leq \beta.$$

The lower bound uses $\lambda/(1 + \lambda) \geq \lambda/(1 + \rho)$. For the sharper upper bound, insert $V = \alpha U$ in the maximum principle

$$\hat{C} = \max_V \{2f^T V - V^T \mathbb{K} V\}.$$

Optimizing the scalar gives $\alpha = (1 + \beta)^{-1}$ and $\hat{C} \geq C/(1 + \beta)$. The familiar participation-weighted bound $e_C \leq \beta$ follows. This is a load-specific statement: w_m and ε_m are evaluated at the same exact assembled trace.

A uniform local inequality $0 \preceq \hat{S}_m - S_m \preceq \varepsilon_* S_m$ implies $T \preceq \varepsilon_* I$, which recovers the load-independent bound $e_C \leq \varepsilon_*/(1 + \varepsilon_*)$ of Eq. (C.3). Rigid reproduction extends a quotient-space bound to arbitrary local traces.

Now take $H_m(t) = tH_{0,m}$ and hold the reference matrices and maps fixed. Set $\mathbb{D}_0 = \sum_m B_m^T H_{0,m}^T A_m H_{0,m} B_m$. Expanding the inverse at zero yields

$$\begin{aligned}\hat{\mathbb{K}}_t &= \mathbb{K} + t^2 \mathbb{D}_0, \\ \hat{U}_t - U &= -t^2 \mathbb{K}^{-1} \mathbb{D}_0 U + O(t^4), \\ C - \hat{C}_t &= t^2 U^T \mathbb{D}_0 U + O(t^4), \\ \hat{u}_{m,t} - u_m &= t J_{I,m}^T H_{0,m} q_m - t^2 E_m B_m \mathbb{K}^{-1} \mathbb{D}_0 U + O(t^3).\end{aligned}\tag{J.2}$$

Thus the retained displacement and compliance errors are $O(t^2)$, the local full-field error is generically $O(t)$, and the retained-error contribution in Eq. (11) is $O(t^4)$. If $\mathbb{D}_0 U = 0$, then $H_{0,m} q_m = 0$ for every affected cell, and this load is reproduced exactly for all t . Uniform estimates across a family of geometries require uniform coercivity and bounded maps; fixed-geometry big-O constants do not supply those estimates automatically.

Appendix J.4. From energy to eight-corner sensitivity

For a fixed retained q , set $u = Eq$, $d = J_I^T H q$, and $\mathcal{E} = d^T K d = (Hq)^T A (Hq)$. Let $D_c = K_{,c}$, $c = 1, \dots, 8$, be symmetric. Define the reference vector $s_c = -u^T D_c u$, and assume $\|s\|_2 > 0$. Eq. (13) then reads

$$\tilde{s}_c - s_c = -2d^T D_c u - d^T D_c d.$$

Let $b_c = J_I D_c u$, $D_{II,c} = J_I D_c J_I^T$, and let the rows of $\mathbf{B}_s \in \mathbb{R}^{8 \times i}$ be $b_c^T A^{-1/2}$. Define

$$L = \|\mathbf{B}_s\|_2, \quad Q = \left(\sum_{c=1}^8 \|A^{-1/2} D_{II,c} A^{-1/2}\|_2^2 \right)^{1/2}.$$

Cauchy–Schwarz and the quadratic-form operator bound give

$$\boxed{\|\tilde{s} - s\|_2 \leq 2L\sqrt{\mathcal{E}} + Q\mathcal{E}, \quad e_s \leq \frac{2L\sqrt{q^T S q}}{\|s\|_2} \sqrt{\varepsilon} + \frac{Qq^T S q}{\|s\|_2} \varepsilon.} \tag{J.3}$$

All eight components are included, and the denominator is the norm of the exact eight-vector for that particular direction. The constants distinguish derivative coupling, interior coercivity, and reference sensitivity scale.

For fixed q and $H = tH_0$, the linear coefficient vanishes precisely when $(H_0q)^T b_c = 0$ for every corner. It vanishes for all interior errors if $J_I D_c E q = 0$ for every corner. A useful special case is $D_c = \alpha_c K$: internal stationarity removes the linear term and the relative sensitivity error equals the energy excess when the reference vector is nonzero.

For a re-equilibrated assembly, the quadratic expansion remains exact with $d_m = \hat{u}_m - u_m$, but d_m now includes a changed retained trace. Eq. (J.2) gives, componentwise,

$$\begin{aligned} \tilde{s}_{m,c}(t) - s_{m,c} = & -2t(J_I^T H_{0,m} q_m)^T D_c u_m \\ & + t^2 \{2(E_m B_m \mathbb{K}^{-1} \mathbb{D}_0 U)^T D_c u_m \\ & - (J_I^T H_{0,m} q_m)^T D_c (J_I^T H_{0,m} q_m)\} + O(t^3). \end{aligned}$$

The leading term is the same as the fixed-trace term. The solution-only replacement is $O(t^2)$; its coefficient can be large on a soft assembly or relative to a small local reference. If only one cell has a learned extension, an exact neighbouring cell has no $O(t)$ reconstruction term, although its solved trace can change at order t^2 .

There is also a global-to-local bound. Suppose K_m has its fixed rigid kernel, each $D_{m,c}$ annihilates that kernel, and

$$\Gamma_m = \left(\sum_c \|K_{m,*}^{-1/2} D_{m,c,*} K_{m,*}^{-1/2}\|_2^2 \right)^{1/2} < \infty$$

on its rigid complement. Set $E_m^{\text{loc}} = u_m^T K_m u_m = w_m C > 0$, $\chi_m = \|\mathbf{s}_m\|_2 / E_m^{\text{loc}} > 0$, and $e_C = (C - \hat{C})/C$. The field expansion and Eq. (11) yield

$$\boxed{e_{s,m} \leq \frac{\Gamma_m}{\chi_m} \left(2\sqrt{\frac{e_C}{w_m}} + \frac{e_C}{w_m} \right)}. \quad (\text{J.4})$$

Indeed, the local error energy is at most $e_C C$. Substitution into $\|\delta \mathbf{s}_m\|_2 \leq \Gamma_m (2\sqrt{E_m^{\text{loc}} a_m(d_m, d_m)} + a_m(d_m, d_m))$ proves the result. This bound makes the roles of small participation and small reference sensitivity explicit. The frozen TPMS records do not measure Γ_m , χ_m , or the local error energy, so Eq. (J.4) is a quantitative theoretical explanation, not a fitted attribution of a reported percentage.

Replacement diagnostics are not an additive sensitivity decomposition. Denote the three terms of Eq. (H.2) by $a = (F - E)q$, $b = E(\hat{q} - q)$, and $c = (F - E)(\hat{q} - q)$, so that the full field error is $a + b + c$. In addition to the field-only and solution-only vector differences, its sensitivity discrepancy contains $-2c^T D_c u - c^T D_c c - 2a^T D_c b - 2a^T D_c c - 2b^T D_c c$. Knowing only the three output norms does not determine these terms or their angles.

Appendix J.5. The complete design derivative and its order

Differentiating $J_I K E = 0$ at fixed coordinates, with $J_P E_{,c} = 0$, gives

$$A E_{I,c} = -J_I K_{,c} E,$$

while the condensed derivative $S_{,c} = E^T K_{,c} E$ is Eq. (H.1). Appendix H also derives the exact assembly compliance derivative $C_{,c} = -U^T \mathbb{K}_{,c} U$ and, from the surrogate's own equilibrium, Eq. (14): $\hat{C}_{,c}$ is the sum over affected cells of $\tilde{s}_{m,c} - 2(F_{I,m,c} \hat{q}_m)^T r_{I,m}$. The design derivative of the solved trace has already been eliminated using equilibrium; the residual term differentiates F at fixed trace. It includes geometry-conditioned coefficients and every design-dependent correction operation.

Equivalently, with $A_{,c} = (K_{,c})_{II}$, differentiate the variational stiffness error:

$$\boxed{(\hat{S} - S)_{,c} = H_{,c}^T A H + H^T A_{,c} H + H^T A H_{,c}.} \quad (\text{J.5})$$

Its norm is bounded by $2\|A\|\|H\|\|H_{,c}\| + \|A_{,c}\|\|H\|^2$. Thus a value error $H = O(t)$ by itself does not imply a quadratic derivative error. A uniformly C^1 -small extension error, $H = tH_0(\tau)$ with bounded $H_0, H_{0,c}$, gives $(\hat{S} - S)_{,c} = O(t^2)$. With uniformly bounded exact solutions and inverse stiffnesses, this also yields $\hat{C}_{,c} - C_{,c} = O(t^2)$. One useful exact splitting is

$$\hat{C}_{,c} - C_{,c} = -\hat{U}^T \Delta_{,c} \hat{U} - (\hat{U} - U)^T \mathbb{K}_{,c} (\hat{U} + U).$$

The $O(t)$ field-estimate term can cancel against the residual term. At the same trace, $J_I K_{,c} E q = -A E_{I,c} q$. For $H = tH_0$, the linear field-estimate error is $+2t(H_0 q)^T A E_{I,c} q$, while the linear residual-chain contribution is $-2t(E_{I,c} q)^T A H_0 q$. This cancellation explains how a first-order field estimate and a second-order complete derivative can coexist.

Under a fixed basis and exact integration, increasing one band parameter τ_c with nonnegative Q_1 shape functions enlarges the material domain. With

fixed ghost stabilisation, this gives $K_{,c} \succeq 0$, $S_{,c} \succeq 0$, and $C_{,c} \leq 0$. The field estimate is then also nonpositive for any field. Pointwise variational stiffness dominance does not itself enforce monotonicity of the surrogate with respect to design: the residual-chain term can change the sign of its complete derivative. This is a separate mechanical consistency question from symmetry and positive definiteness at a fixed design. Numerical moment derivatives inherit the monotonicity statement only when they represent the same nested-domain integration rule.

More explicitly, for $h > 0$, $N_c^{Q_1}(x) \geq 0$ implies $\Omega(\tau) \subseteq \Omega(\tau + he_c)$. For any discrete coefficient vector v ,

$$v^T [K(\tau + he_c) - K(\tau)]v = \int_{\Omega(\tau + he_c) \setminus \Omega(\tau)} \epsilon(v) : \mathbb{C} : \epsilon(v) dx \geq 0. \quad (\text{J.6})$$

Here the material tensor and basis are fixed and the ghost contribution cancels. Taking the differentiable limit proves positive semidefiniteness of the thickness derivative. A centred difference of stiffness matrices assembled from exactly nested domains is also PSD. Changes of adaptive integration subdivision, approximate moments, or independently selected stabilisation can interrupt that discrete nesting relation; fixed active topology alone does not verify its numerical preservation. Section 6.2 and Figure S01(c) report a step-refinement study of the numerical derivative on four cells: relative to the production step $10^{-5}\tau_c$, the sensitivity changes by at most 2.6×10^{-7} at $10^{-3}\tau_c$, 2.5×10^{-9} at $10^{-4}\tau_c$ and 1.6×10^{-10} at $10^{-6}\tau_c$; the hundredfold reduction per decade is the second-order truncation of the central difference, so no integration branch changes within $\pm 10^{-3}\tau_c$ at the fixed active set on these cells. The element-level check of Supplementary Table ST15 shows that the discrete moments do not preserve the nesting everywhere. The element stiffnesses are positive semidefinite to rounding, but the exact derivative of the discrete moments at fixed clipping topology gives element derivative matrices with negative eigenvalues: in the uncut cells the smallest ratio $\lambda_{\min}/\max|\lambda|$ is -2.3×10^{-3} , and in four of the six cut cells between 2 and 11 partially filled elements (of 762 to 6,033) have a uniform-thickening derivative that is negative semidefinite. The production central difference agrees with this exact discrete derivative to 2×10^{-8} . Eq. (J.6), and the bound (J.7) that rests on it, therefore describe exact integration; the discrete model satisfies them only approximately. The reference sensitivities of this study are derivatives of the discrete model and are compared as such.

This positive structure strengthens the sensitivity interpretation with-

out requiring an indefinite derivative. Put $a_c = u^T K_{,c} u = -s_c \geq 0$, $\zeta_c = d^T K_{,c} d \geq 0$. Positive-semidefinite Cauchy–Schwarz gives

$$|\tilde{s}_c - s_c| \leq 2\sqrt{a_c \zeta_c} + \zeta_c, \quad \|\tilde{\mathbf{s}} - \mathbf{s}\|_2 \leq 2\sqrt{\sum_c a_c \zeta_c} + \|\zeta\|_2. \quad (\text{J.7})$$

If $\gamma_c = \|A^{-1/2} A_{,c} A^{-1/2}\|_2$, then $\zeta_c \leq \gamma_c \mathcal{E}$, giving $2\sqrt{\sum_c a_c \gamma_c} \sqrt{\mathcal{E}} + \|\gamma\|_2 \mathcal{E}$. The quadratic term in the signed sensitivity discrepancy is nonpositive; the cross term can have either sign. The reference eight-vector has nonpositive components. A small reference norm reflects small derivative-weighted local strain energy. For a general variable that moves a cut or redistributes material, an indefinite derivative is possible; that generality is unnecessary to explain the present corner-thickening mechanism.

Design-dependent loads and assembly maps add the chain-rule terms stated after Eq. (H.5). These changes are different derivative problems rather than modifications of Eq. (J.5).

Appendix J.6. Finite iteration residual and energy/action consistency

Let $\bar{U}$ be an approximate solution of the same fixed variational operator, $\rho = f - \hat{\mathbb{K}}\bar{U}$, $\bar{C} = f^T \bar{U}$, and $\bar{u}_m = F_m B_m \bar{U}$. The exact identity becomes

$$\boxed{C - \bar{C} = a(\bar{u} - u, \bar{u} - u) + \bar{U}^T \rho, \quad J(\bar{U}) = \bar{C} + \bar{U}^T \rho \leq \hat{C} \leq C.}$$

The error of the corrected functional is $\hat{C} - J(\bar{U}) = \rho^T \hat{\mathbb{K}}^{-1} \rho$, and $\hat{C} - \bar{C} = \bar{U}^T \rho$. Thus the work $f^T \bar{U}$ can violate the variational compliance ordering if the algebraic residual is appreciable, while the concave work functional retains a quadratic residual error. A Euclidean residual bound converts through

$$\frac{|\bar{U}^T \rho|}{C} \leq \frac{\|\bar{U}\|_2 \|f\|_2 \|\rho\|_2}{C \|f\|_2},$$

with a load- and stiffness-dependent constant.

If an applied action $y(\bar{U})$ is not numerically identical to the variational energy, define $\omega = \bar{U}^T y(\bar{U}) - \sum_m \bar{u}_m^T K_m \bar{u}_m$ and $\rho = f - y(\bar{U})$. Then the first identity acquires an additional $+\omega$. This separates solve residual from action/energy inconsistency. A recursive Krylov residual need not equal this applied-action residual. Evaluating the terms in Eq. (18) additionally requires the signed residual work and ω . Instrumented re-solves of the pair

configurations and lattices record both, together with the dual-norm bound $|\bar{U}^T \rho| \leq \sqrt{\bar{U}^T y(\bar{U})} \sqrt{\rho^T \mathbb{K}^{-1} \rho}$, which holds because $\hat{\mathbb{S}} \succeq \mathbb{S}$ (Supplementary Note S6.3): $|\bar{U}^T \rho|$ and $|\omega|$ stay below 6×10^{-8} of the compliance and the bound below 2×10^{-5} , so the reported errors are those of the operator.

Appendix J.7. Directional training, coverage, rotations, and spectra

With Z and $S_* = Z^T S Z \succ 0$ as in Eq. (B.5), let $T_* = S_*^{-1/2} Z^T H^T A H Z S_*^{-1/2}$. For an energy-normalised direction, $x_j = S_*^{1/2} Z^T q_j$ has unit Euclidean norm and, by Eq. (B.6), $\varepsilon_j = x_j^T T_* x_j$. Define its coverage matrix $M_N = N^{-1} \sum_j x_j x_j^T$. Then

$$\bar{\varepsilon} = \text{tr}(T_* M_N), \quad M_N \succeq \alpha I, \alpha > 0 \implies \varepsilon_* \leq \text{tr}(T_*) \leq \bar{\varepsilon} / \alpha. \quad (\text{J.8})$$

If the directions do not span the complement, choose a unit v orthogonal to them and $T_* = M v v^T$. Every sampled error is zero and the worst error is M , arbitrarily large. Such a PSD perturbation can be realised as $H^T A H$ if there is at least one interior degree of freedom. Thus retaining all coordinates in $\mathbf{P}$ defines the approximation target, while adequate direction coverage controls its learned accuracy.

The logarithmic loss is nonnegative for the exact variational construction and behaves as $\log(1 + \varepsilon) = \varepsilon + O(\varepsilon^2)$ near zero. It weights large directional errors differently from their arithmetic mean. A finite average log loss L_N only supplies the weak sample bound $\sum_j \varepsilon_j \leq e^{N L_N} - 1$; it has no unsampled-direction implication without coverage. With exact reference normalisation, Ritz searches give attained Rayleigh quotients and hence lower bounds on the true largest quotient. Geometry-averaged direction means, their population maxima, and the full operator supremum are different statistics.

For an orthogonal signed-permutation action, the transformations in Eq. (G.2) together with $q' = T_P q$ and $B'_m = T_P B_m T_g^T$ preserve energies, nullspaces and assembly. A correspondingly transformed extension has the same property. Training on transformed examples encourages this relation but does not prove that an unconstrained learned map satisfies it on every group element. Geometry and loading transform together, and fields are returned to the original frame before comparison.

The identity $\varepsilon = \delta^2 \kappa$ in Eq. (B.8) uses the same displacement weighting in both Rayleigh quotients. Its denominator measures the physical response in the chosen trace direction. The smoother instead acts on the spectrum

of $Av = \lambda Dv$, $D = \text{diag } A$, with trace fixed. These concepts of softness are distinct. For example, $A = \text{diag}(\epsilon, 1)$ has a very soft unscaled direction, while $D^{-1/2}AD^{-1/2} = I$. An internal eigenmode also has zero retained trace and is not itself a free-cell Schur response. Consequently, cumulative Jacobi-scaled low-mode error energy identifies slowly relaxed components; it does not determine the measured κ , a continuum bending-mode label, or a model-capacity lower bound.

Appendix J.8. Corrections preserve the target and can improve different quantities differently

A geometry-fixed linear correction acts on H through $H_{\text{new}} = TH$. If $T^T AT \preceq A$, then

$$S \preceq \hat{S}_{\text{new}} \preceq \hat{S}_{\text{old}}.$$

Thus exact assembled compliance increases monotonically toward the reference, and Eq. (11)'s total reconstructed energy error decreases. Neither a signed component of displacement nor the field-based sensitivity norm is ordered by this matrix inequality.

Appendix D establishes the two components of such a correction: Chebyshev smoothing satisfies the energy bound of Eq. (D.2), with $\rho_k \leq 1$ when the actual positive spectrum lies in $(0, b]$, and the exact Galerkin update $C_V = I - VA_c^{-1}V^T A$, $A_c = V^T AV$, is the A -orthogonal projection of Eq. (D.5). For a symmetric approximate inverse G in place of A_c^{-1} , the precise condition is

$$\|d\|_A^2 - \|d - VGV^T Ad\|_A^2 = b_r^T (2G - GA_c G) b_r, \quad b_r = V^T Ad. \quad (\text{J.9})$$

Consequently $2G - GA_c G \succeq 0$ suffices for nonexpansiveness. For an accurate A_c , a nonnegative shift $G = (A_c + \Lambda)^{-1}$, $\Lambda \succeq 0$, satisfies this condition, although it is not the exact projection. This statement concerns the coarse inverse while the prescribed fine-grid stiffness K remains fixed. If a probed matrix differs from A_c , contraction depends on the approximate inverse relative to the true A_c .

The pre-smooth/coarse/post-smooth cycle $H_{\text{tg}} = P_k C_V P_k H_0$ then satisfies the ρ_k^4 energy bound derived in Appendix D. At the operator-application level the entire affine field update, including the retained forcing, is transposed in reverse order, as in Eqs. (E.3)–(E.4). Geometry-fixed coefficients

and a fixed iteration count ensure linearity. A fixed number of ordinary CG iterations generally does not: its coefficients depend on the right-hand side. Arbitrary direction-dependent stopping also need not define one linear extension.

The benefit of learning under a fixed correction budget is measured by the error surviving the correction: $\|TH_{\text{net}}q\|_A^2$ versus $\|TH_{\text{start}}q\|_A^2$. Initial errors with the same energy can leave different corrected errors. This connects direction-sensitive training to the numerical correction actually used.

Boundary-space restriction and interior correction affect different trial spaces. Exact interior condensation followed by a restriction $U = G_r y$ solves over a subspace of the retained coordinates; nested boundary spaces give nondecreasing Ritz compliance. Interior correction preserves all retained coordinates and changes the interior graph $u = FBU$; its matrix ordering comes from contraction of H , even though two such graph spaces need not be nested. In both cases compliance can approach the reference monotonically while a local sensitivity norm does not.

Appendix J.9. Illustrative matrix examples

The following matrix examples isolate the roles of error amplitude, derivative coupling, and design regularity.

1. **Orders under re-equilibration.** A two-retained/three-internal SPD cell is coupled to a three-coordinate supported assembly. Five halved values of t verify the variational identity, Eq. (11), Eq. (12) and the sensitivity expansion. The saved final log-two slopes are approximately 2 for the operator, retained displacement and compliance, 1 for the full field and field-based sensitivity, and 4 for the retained-error energy in Eq. (11). Supplementary Note S8 gives the matrices, parameter sequence, and saved slope values for this illustrative algebraic example.
2. **Energy decreases while sensitivity error increases, even for a PSD derivative.** At the base design take $K = I_3$, $P = \{1\}$, $u = (1, 0, 0)^T$, $v = (1, 1, -1)^T$, and $K_c = vv^T \succeq 0$. The family $K(\tau) = I + \tau vv^T$ is SPD for $\tau > -1/3$ and increases with τ . Start with error $d = (0, t, t)^T$ and project out the second coordinate. Its energy falls from $2t^2$ to t^2 , while the sensitivity discrepancy changes from zero to $2t - t^2$. The reference sensitivity is -1 , so this is also a relative-error counterexample. At $t = .1$, the energies are .02 and .01 and the sensitivity discrepancies are zero and .19. The construction has the positive derivative structure of a nested thickening family.

3. **First-order field estimate, second-order total derivative with a PSD stiffness derivative.** Let

$$K(\tau) = \begin{bmatrix} 1+\tau & \tau \\ \tau & 1+\tau \end{bmatrix}, \quad E = (1, -\tau/(1+\tau))^T, \quad F_t = E + (0, t)^T.$$

Then $S = (1+2\tau)/(1+\tau)$ and $\hat{S} = S + (1+\tau)t^2$. For $f = 1$ at $\tau = 0$, the exact derivative is -1 , the field estimate is $-(1+t)^2/(1+t^2)^2$, and the residual-chain term is $2t/(1+t^2)^2$. Their sum is $-1/(1+t^2)$, whose error is $t^2/(1+t^2)$. The $O(t)$ terms cancel while the stiffness derivative is positive semidefinite.

4. **Small values without small derivatives.** For a fixed load $f = 1$, let $K = I_2$, $E = (1, 0)^T$, and $F_t(\tau) = (1, t+\tau)^T$. At $\tau = 0$, the compliance gap is $C - \hat{C}_t = t^2/(1+t^2)$, whereas the signed derivative error is

$$\partial_\tau(\hat{C}_t - C)\big|_{\tau=0} = -\frac{2t}{(1+t^2)^2}.$$

Uniform value convergence also does not imply derivative convergence: take $H_t(\tau) = t \sin(\tau/t^2 + \pi/4)$. At $\tau = 0$, the compliance gap is $O(t^2)$, but

$$\partial_\tau(\hat{C}_t - C)\big|_{\tau=0} = -\frac{1}{(1+t^2/2)^2} \rightarrow -1.$$

Each family is differentiable; neither is uniformly C^1 -small. The derivative of the positive gap $C - \hat{C}_t$ has the opposite sign.

5. **Fixed-design structure does not imply design monotonicity.** For $K(c) = cI_2$, $c > 0$, $E = (1, 0)^T$, and $F = (1, h(c))^T$, one has $\hat{S} = c(1+h^2) \geq S = c$. At $c = 1$, $h = .1$, $h' = -10$, $\hat{S}' = -.99$ and $\hat{C}' > 0$ for $f = 1$, although $C' = -1$. The field estimate remains negative. This counterexample isolates design variation of the extension from its valid pointwise energy structure.
6. **Trace drift and normalisation.** If an implemented field has retained coordinates $q + b$, then splitting the term $d^T K d$ in Eq. (F.2), with $b_P = b$, gives, relative to the original input,

$$\hat{u}^T K \hat{u} - q^T S q = 2b^T S q + b^T S b + \|\hat{u}_I - E_I(q + b)\|_A^2. \quad (\text{J.10})$$

Thus a first-order trace or normalisation discrepancy can mask a small quadratic internal error. Relative to the actual retained vector, the

nonnegative internal-energy identity remains the appropriate comparison. Equation (J.10) separates retained-coordinate drift from internal reconstruction error and specifies the quantities needed for a consistent energy comparison.

References

- Adams, M., Brezina, M., Hu, J., Tuminaro, R. (2003). Parallel multigrid smoothing: polynomial versus Gauss–Seidel. *Journal of Computational Physics*, 188(2), 593–610.
- Alexandersen, J., Lazarov, B. S. (2015). Topology optimisation of manufacturable microstructural details without length scale separation using a spectral coarse basis preconditioner. *Computer Methods in Applied Mechanics and Engineering*, 290, 156–182.
- Amir, O., Bendsøe, M. P., Sigmund, O. (2009). Approximate reanalysis in topology optimization. *International Journal for Numerical Methods in Engineering*, 78(12), 1474–1491.
- Amir, O., Stolpe, M., Sigmund, O. (2010). Efficient use of iterative solvers in nested topology optimization. *Structural and Multidisciplinary Optimization*, 42(1), 55–72.
- Ballani, J., Huynh, D. B. P., Knezevic, D. J., Nguyen, L., Patera, A. T. (2018). A component-based hybrid reduced basis/finite element method for solid mechanics with local nonlinearities. *Computer Methods in Applied Mechanics and Engineering*, 329, 498–531.
- Becker, R., Rannacher, R. (2001). An optimal control approach to a posteriori error estimation in finite element methods. *Acta Numerica*, 10, 1–102.
- Bendsøe, M. P., Sigmund, O. (2003). *Topology Optimization: Theory, Methods, and Applications*, 2nd edn. Springer, Berlin, Heidelberg.
- Bonilla Moreno, G., Guarino, G., Antolin, P. (2027). A ROM-based BDDC solver for unfitted p-FEM level-set-based two-dimensional lattice structures. *Computer Methods in Applied Mechanics and Engineering*, 463, 119304.
- Boullé, N., Townsend, A. (2023). Learning elliptic partial differential equations with randomized linear algebra. *Foundations of Computational Mathematics*, 23(2), 709–739.
- Burman, E., Claus, S., Hansbo, P., Larson, M. G., Massing, A. (2015). CutFEM: Discretizing geometry and partial differential equations. *International Journal for Numerical Methods in Engineering*, 104(7), 472–501.

- Chen, W., Li, M. (2026). Conditioned numerical shape functions on unfitted reduced coarse elements for robust analysis of complex solid structures. *Computer-Aided Design*, 193, 104038.
- Dohrmann, C. R. (2003). A preconditioner for substructuring based on constrained energy minimization. *SIAM Journal on Scientific Computing*, 25(1), 246–258.
- Dohrmann, C. R. (2007). An approximate BDDC preconditioner. *Numerical Linear Algebra with Applications*, 14(2), 149–168.
- E, W., Yu, B. (2018). The Deep Ritz method: a deep learning-based numerical algorithm for solving variational problems. *Communications in Mathematics and Statistics*, 6(1), 1–12.
- Efendiev, Y., Galvis, J., Hou, T. Y. (2013). Generalized multiscale finite element methods (GMsFEM). *Journal of Computational Physics*, 251, 116–135.
- Eftang, J. L., Patera, A. T. (2013). Port reduction in parametrized component static condensation: approximation and a posteriori error estimation. *International Journal for Numerical Methods in Engineering*, 96(5), 269–302.
- Falgout, R. D., Vassilevski, P. S., Zikatanov, L. T. (2005). On two-grid convergence estimates. *Numerical Linear Algebra with Applications*, 12(5–6), 471–494.
- Falgout, R. D., Yang, U. M. (2002). hypre: A library of high performance preconditioners. In *Computational Science — ICCS 2002*, Lecture Notes in Computer Science, 2331, 632–641. Springer, Berlin, Heidelberg.
- Farhat, C., Lesoinne, M., LeTallec, P., Pierson, K., Rixen, D. (2001). FETI-DP: a dual–primal unified FETI method—part I: A faster alternative to the two-level FETI method. *International Journal for Numerical Methods in Engineering*, 50(7), 1523–1544.
- Fraeijs de Veubeke, B. (1965). Displacement and equilibrium models in the finite element method. In Zienkiewicz, O. C., Holister, G. S. (eds.), *Stress Analysis*, Chapter 9, 145–197. John Wiley & Sons. Reprinted in *International Journal for Numerical Methods in Engineering*, 52(3) (2001), 287–342, doi:10.1002/nme.339.
- Giles, M. B., Pierce, N. A. (2000). An Introduction to the Adjoint Approach to Design. *Flow, Turbulence and Combustion*, 65(3–4), 393–415.
- Gogu, C. (2015). Improving the efficiency of large scale topology optimization through on-the-fly reduced order model construction. *International Journal for Numerical Methods in Engineering*, 101(4), 281–304.
- Golub, G. H., Varga, R. S. (1961). Chebyshev semi-iterative methods, successive overrelaxation iterative methods, and second order Richardson iterative methods. Part I. *Numerische Mathematik*, 3(1), 147–156.

- Greenfeld, D., Galun, M., Basri, R., Yavneh, I., Kimmel, R. (2019). Learning to optimize multigrid PDE solvers. In *Proceedings of the 36th International Conference on Machine Learning*, PMLR 97, 2415–2423.
- Guo, Y., Liu, C., Du, Z., Jia, Y., Jiang, C., Guo, X., Shen, C. (2026a). High-Generalization AI-Enhanced Mechanical Analysis and Topology Optimization via Cubic Bézier Interpolation of Substructure Boundary Displacements. *Computer Methods in Applied Mechanics and Engineering*, 456, 118955.
- Guo, Y., Liu, C., Du, Z., Liu, J., Feng, J., Zhang, X., Li, Y., Yang, T., Shen, C., Guo, X. (2026b). PIML-OFEM: A New Large-Scale Structural Analysis Method Based on Problem-Independent Machine Learning and Overlapping Finite Element Technique. arXiv:2607.22019v1 [Preprint].
- Guyan, R. J. (1965). Reduction of stiffness and mass matrices. *AIAA Journal*, 3(2), 380.
- Hackbusch, W. (1985). *Multi-Grid Methods and Applications*. Springer, Berlin.
- Haftka, R. T., Gürdal, Z. (1992). *Elements of Structural Optimization*, 3rd edn. Kluwer, Dordrecht.
- He, J., Liu, X., Xu, J. (2024). MgNO: Efficient parameterization of linear operators via multigrid. In *International Conference on Learning Representations (ICLR 2024)*, 53409–53428. OpenReview: 8OxL034uEr.
- Heinlein, A., Klawonn, A., Lanser, M., Weber, J. (2021). Combining machine learning and adaptive coarse spaces—a hybrid approach for robust FETI-DP methods in three dimensions. *SIAM Journal on Scientific Computing*, 43(5), S816–S838.
- Henson, V. E., Yang, U. M. (2002). BoomerAMG: A parallel algebraic multigrid solver and preconditioner. *Applied Numerical Mathematics*, 41(1), 155–177.
- Hou, T. Y., Wu, X.-H. (1997). A Multiscale Finite Element Method for Elliptic Problems in Composite Materials and Porous Media. *Journal of Computational Physics*, 134(1), 169–189.
- Huang, M., Du, Z., Liu, C., Zheng, Y., Cui, T., Mei, Y., Li, X., Zhang, X., Guo, X. (2022). Problem-independent machine learning (PIML)-based topology optimization—A universal approach. *Extreme Mechanics Letters*, 56, 101887.
- Huang, M., Cui, T., Liu, C., Du, Z., Zhang, J., He, C., Guo, X. (2023). A Problem-Independent Machine Learning (PIML) enhanced substructure-based approach for large-scale structural analysis and topology optimization of linear elastic structures. *Extreme Mechanics Letters*, 63, 102041.
- Huang, M., Liu, C., Guo, Y., Zhang, L., Du, Z., Guo, X. (2024). A mechanics-based data-free Problem Independent Machine Learning (PIML) model for

- large-scale structural analysis and design optimization. *Journal of the Mechanics and Physics of Solids*, 193, 105893.
- Huynh, D. B. P., Knezevic, D. J., Patera, A. T. (2013). A static condensation reduced basis element method: approximation and a posteriori error estimation. *ESAIM: Mathematical Modelling and Numerical Analysis*, 47(1), 213–251.
- Irons, B. (1965). Structural eigenvalue problems—elimination of unwanted variables. *AIAA Journal*, 3(5), 961–962.
- Jiang, C., Liu, C., Guo, Y., Du, Z., Zhang, W., Guo, X. (2026). Mechanically consistent equivariant machine learning for substructural analysis. *Composite Structures*, 397, 120865.
- Jiang, H., Zhan, J., Zhang, C., Wang, F. (2026). Convex Neural Energy Elements: Monolithic Finite-Element Assembly of Geometry-Parameterized Neural Operators with Stability and Error Guarantees. arXiv:2608.02036v1 [Preprint].
- Klawonn, A., Rheinbach, O. (2007). Inexact FETI-DP methods. *International Journal for Numerical Methods in Engineering*, 69(2), 284–307.
- Kopaničáková, A., Karniadakis, G. E. (2025). DeepONet Based Preconditioning Strategies for Solving Parametric Linear Systems of Equations. *SIAM Journal on Scientific Computing*, 47(1), C151–C181.
- Li, J., Widlund, O. B. (2007). On the use of inexact subdomain solvers for BDDC algorithms. *Computer Methods in Applied Mechanics and Engineering*, 196(8), 1415–1428.
- Li, Z., Huang, D. Z., Liu, B., Anandkumar, A. (2023a). Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine Learning Research*, 24(388), 1–26.
- Li, Z., Kovachki, N., Choy, C., Li, B., Kossaiji, J., Otta, S., Nabian, M. A., Stadler, M., Hundt, C., Azizzadenesheli, K., Anandkumar, A. (2023b). Geometry-informed neural operator for large-scale 3D PDEs. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 35836–35854.
- Luz, I., Galun, M., Maron, H., Basri, R., Yavneh, I. (2020). Learning algebraic multigrid using graph neural networks. In *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119, 6489–6499.
- Oden, J. T., Prudhomme, S. (2001). Goal-oriented error estimation and adaptivity for the finite element method. *Computers & Mathematics with Applications*, 41(5–6), 735–756.
- Parish, E., Lindsay, P., Shelton, T., Mersch, J. (2024). Embedded symmetric positive semi-definite machine-learned elements for reduced-order modeling in finite-element simulations with application to threaded fasteners. *Computa-*

- tional Mechanics*, 74(6), 1357–1381.
- Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., Battaglia, P. W. (2021). Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations (ICLR 2021)*.
- Smetana, K., Patera, A. T. (2016). Optimal Local Approximation Spaces for Component-Based Static Condensation Procedures. *SIAM Journal on Scientific Computing*, 38(5), A3318–A3356.
- Toselli, A., Widlund, O. (2005). *Domain Decomposition Methods — Algorithms and Theory*. Springer, Berlin.
- Trottenberg, U., Oosterlee, C. W., Schüller, A. (2001). *Multigrid*. Academic Press, San Diego. ISBN 0-12-701070-X.
- Um, K., Brand, R., Fei, Y., Holl, P., Thuerey, N. (2020). Solver-in-the-loop: learning from differentiable physics to interact with iterative PDE-solvers. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 6111–6122.
- Vaněk, P., Mandel, J., Brezina, M. (1996). Algebraic multigrid by smoothed aggregation for second and fourth order elliptic problems. *Computing*, 56(3), 179–196.
- Xing, Y., Liu, Y., Xue, T., Lu, L. (2026). GMT: A Geometric Multigrid Transformer Solver for Microstructure Homogenization. *ACM Transactions on Graphics*, 45(4), 1–15.
- Xu, J. (1992). Iterative Methods by Space Decomposition and Subspace Correction. *SIAM Review*, 34(4), 581–613.
- Xu, J., Zikatanov, L. (2002). The method of alternating projections and the method of subspace corrections in Hilbert space. *Journal of the American Mathematical Society*, 15(3), 573–597.
- Xu, W., Liu, C., Guo, Y., Huang, M., Guo, X. (2025). Problem-Independent Machine Learning (PIML) enhanced 3D lattice composite structures optimization via moving morphable components approach. *Composite Structures*, 369, 119330.
- Yu, S., Sun, J., Bai, J. (2019). Investigation of functionally graded TPMS structures fabricated by additive manufacturing. *Materials & Design*, 182, 108021.
- Zhang, E., Kahana, A., Kopaničáková, A., Turkel, E., Ranade, R., Pathak, J., Karniadakis, G. E. (2024). Blending neural operators and relaxation methods in PDE numerical solvers. *Nature Machine Intelligence*, 6(11), 1303–1313.
- Zhang, L., Liu, C., Guo, Y., Jiang, C., Guo, X. (2026). Problem-independent transfer learning for complex-domain 3D topology optimization. *International*

Journal of Mechanical Sciences, 328, 112007.